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Aluminium outlook

Electricity in the form of metal - sleepwalking into the biggest deficits in over 20 years - Jan 2026 update

Wenyu Yao^{AC}
Metals Strategy
wenyu.yao@citi.com
+44 20 7986 4551

Maximilian Layton^{AC}
Global Head | Commodities
max.layton@citi.com
+44 20 7986 4556

Shreyas Madabushi^{AC}
Metals Strategy
shreyas.madabushi@citi.com
+91 22 4277 5048

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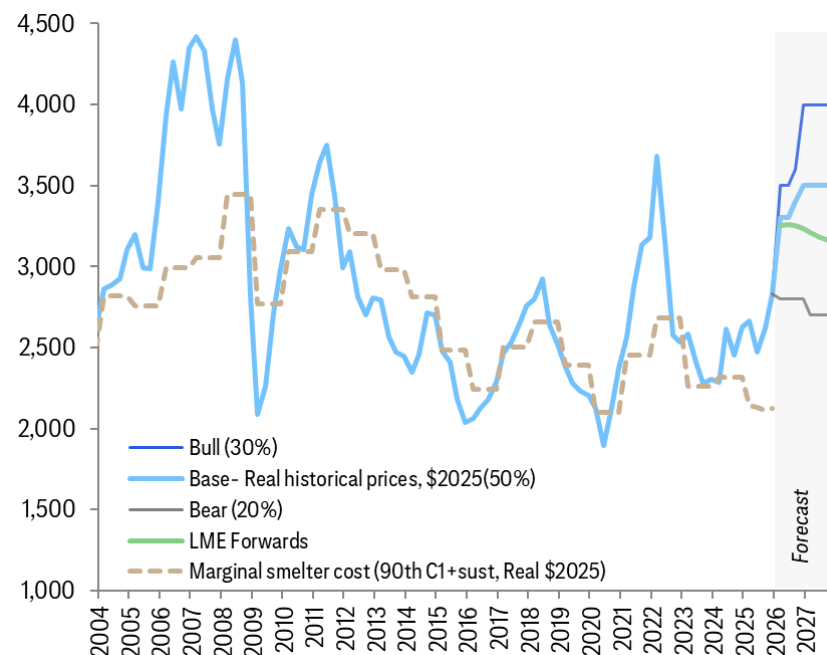
Aluminium – constrained supply gives leverage to the cycle, ~10-20% upside

We remain bullish on aluminium over the medium to long term and strongly recommend consumer hedging at current levels over a 1~5-year horizon. In our view, while prices have risen by 30% since we first turned structurally bullish in June 2025 ([link](#)) and subsequently first published the original version of this bullish aluminium slide deck in September 2025 ([link](#)), they remain insufficient to unlock the level and pace of supply growth required to keep the market balanced over the medium to long term. We estimate that a sustained and credible period meaningfully above US\$3,500/t is likely required to incentivise the >10 Mt of additional supply needed by 2030. Even at these levels, supply growth is likely to be gradual and constrained by global competition for power. At materially higher prices, incremental supply from Indonesia and scrap could partially alleviate projected imbalances. Absent this, the market may ultimately need to clear via demand destruction, potentially at prices approaching US\$4,000/t.

Future facing on the demand side, power constrained on the supply side

- **Robots (humanoid, dogs, drones, industrial robots etc., with 20-25kg aluminium per humanoid robot), AI/datacentres and drones could be massive new demand drivers for aluminium**, especially taking a longer-term view. These demand segments will compete for the very same power that the energy-intensive aluminium smelters need to grow supply.
- **Asset allocators are likely moving to increase the share of aluminium in their portfolios over the coming years** and be joined by strong (macro, commodity and other) fund/investor buying once growth sentiment turns higher in late 2026/2027, in our view. And we saw what that did for copper over the past few years (i.e., how copper boomed during Jan-April 2024).
- **Genuine deficits, stockouts risks, and backwardation are much more likely in aluminium than copper over the coming years**, we believe, since aluminium scrap is far less able to respond to higher prices than copper (due to technical constraints).

Citi aluminium price forecast scenarios



Price Forecasts	Scenario Weight	Unit	0-3 mth pt price	6-12 mth pt price	1Q26f	2Q26f	3Q26f	4Q26f	2026f	1Q27f	2Q27f	3Q27f	4Q27f	2027f	2028f
Aluminium (Bull)	30%	\$/t			3500	3500	3600	4000	3650	4000	4000	4000	4000	4000	4200
Aluminium (Base)	50%	\$/t	3400	3500	3300	3300	3400	3500	3375	3500	3500	3500	3500	3500	3800
Aluminium (Bear)	20%	\$/t			2800	2800	2800	2800	2800	2700	2700	2700	2700	2700	2700

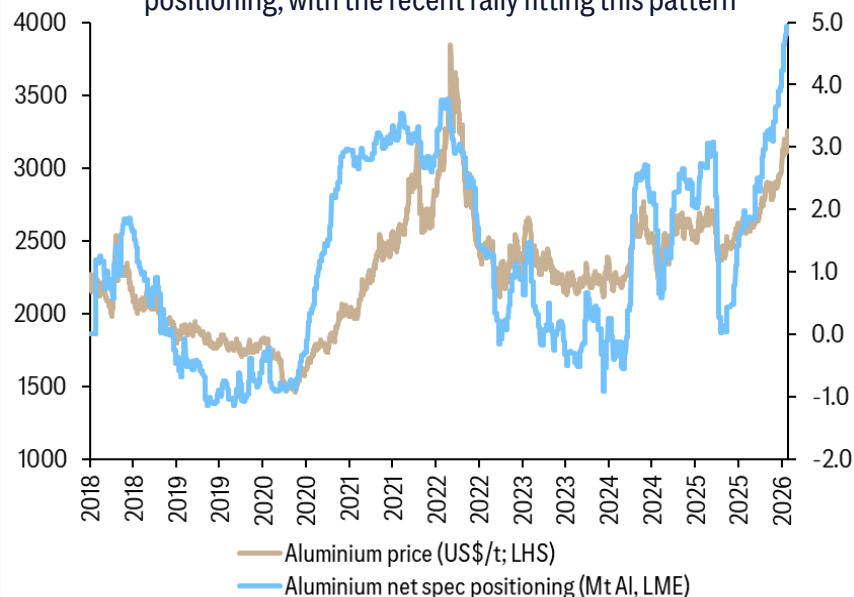
Investors have brought forward AL's structural case as cyclical expectations build

Net speculative length in LME aluminium has risen meaningfully, suggesting investors have moved to price in an expectation of cyclical improvement in demand as the broader macro backdrop turns more supportive. A benign environment — with decent global growth, falling inflation and expectations of continued Fed easing in 2026 - has encouraged renewed risk appetite across cyclicals. This has helped pull positioning ahead of very near-term hard data, leaving sentiment more exposed to incremental macro or activity surprises. Since we first published our structural aluminium bull-case deck (Sep 2025), prices have risen by c.20%, reinforcing investor confidence in the longer-term narrative.

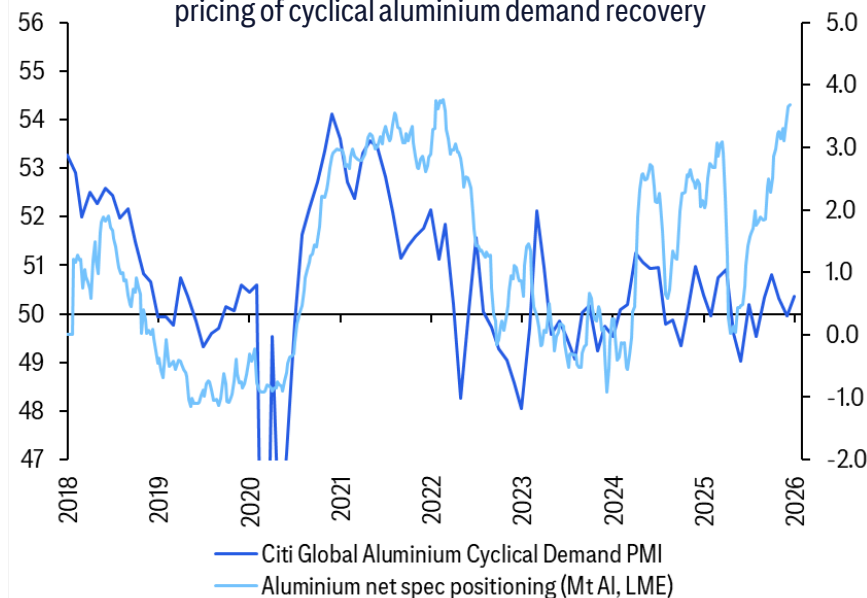
Recent flows also appear to reflect wider macro themes - including concerns around potential currency debasement and renewed interest in hard-asset hedges at a time when no major central bank is close to turning hawkish. Importantly, investors are also leaning into the medium- to long-term structural bull case for aluminium, which adds an additional layer of conviction to positioning. Together with a constructive liquidity backdrop and a preference for base metals within the broader commodities complex, this has effectively brought forward part of the prospective bull case, with expectations embedded earlier than underlying fundamentals alone would imply.

Downside risks could present dip-buying opportunity if the risks materialise. A potential Japan debt-related shock, wider geopolitical uncertainties across multiple regions or a softer global industrial cycle or weaker AI productivity gains, could all challenge the current strong aluminium net speculative positioning. A stronger USD or a tightening in global liquidity would also act as headwinds.

Historically, large aluminium price moves have been dominating by speculative positioning, with the recent rally fitting this pattern

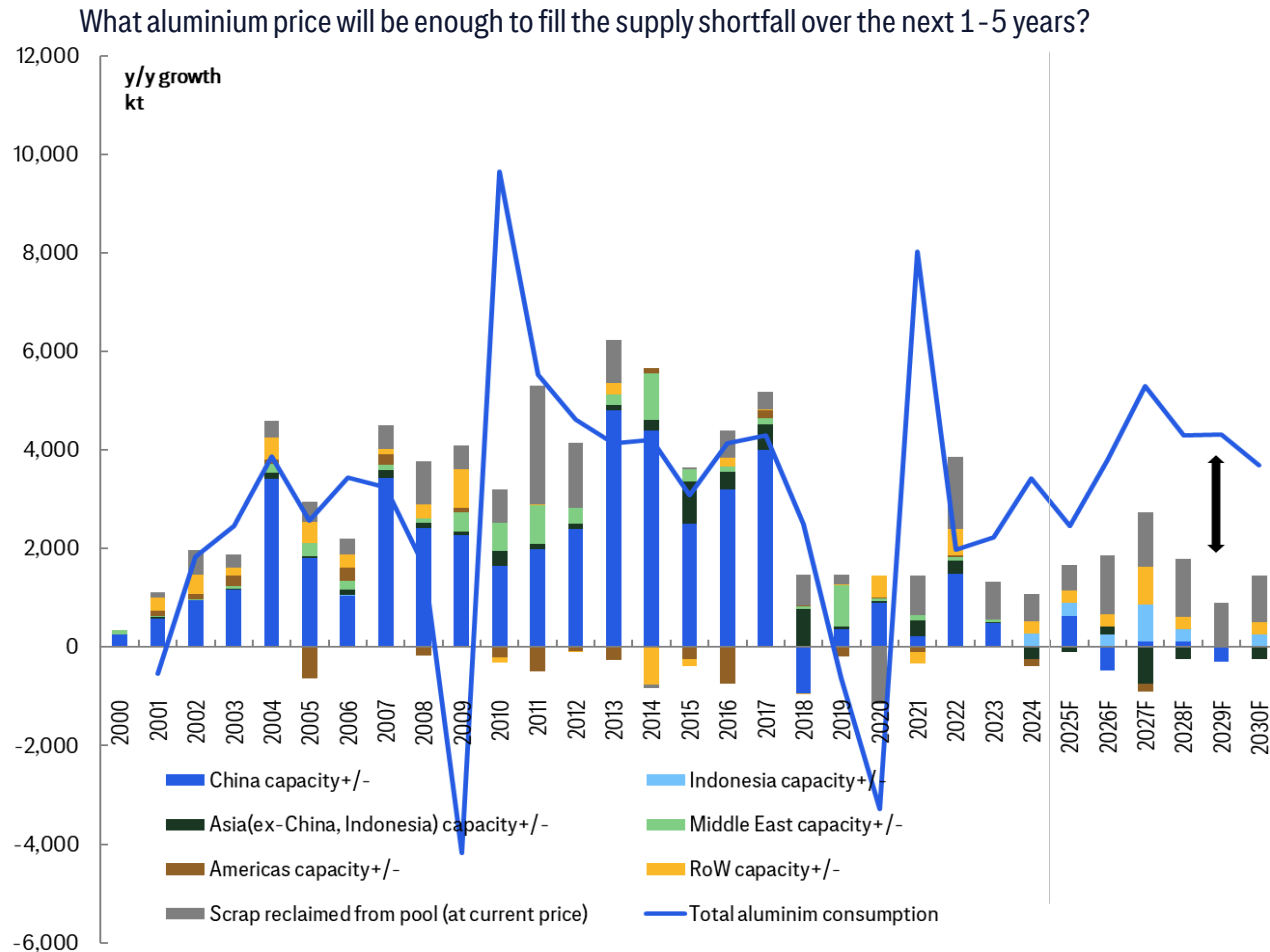


LME aluminium positioning dynamics imply a forward-pricing of cyclical aluminium demand recovery



Aluminium demand growth seems set to outpace supply growth for years to come

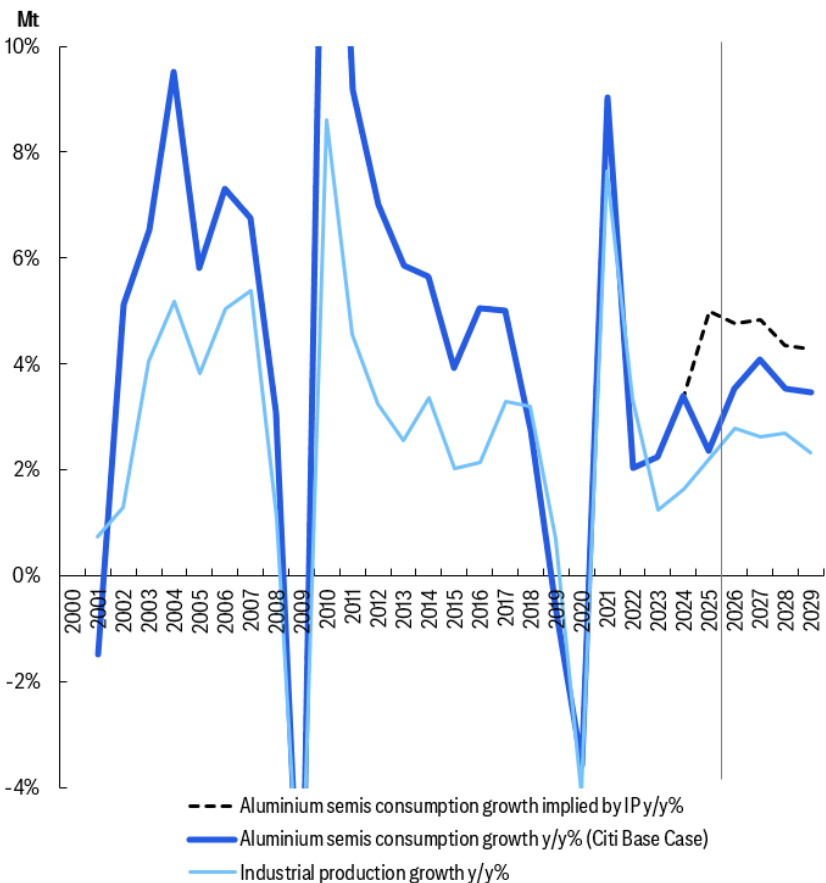
The era of rapid aluminium capacity expansion driving production growth has ended, with primary supply growth now structurally limited. Meanwhile, we expect total aluminium demand to continue rising in the coming years, particularly as cyclical demand recovers. Under our current base-case assumptions for projects and secondary supply, the market appears highly vulnerable to a sizeable deficit, as demand growth is set to outpace supply growth. In our view, the market can be rebalanced over the medium to long term only through materially higher prices (likely well above US\$3,000/t and potentially approaching US\$4,000/t, as required in the early 2000s), surprisingly weak demand, or a technological or legislative breakthrough in scrap recovery.



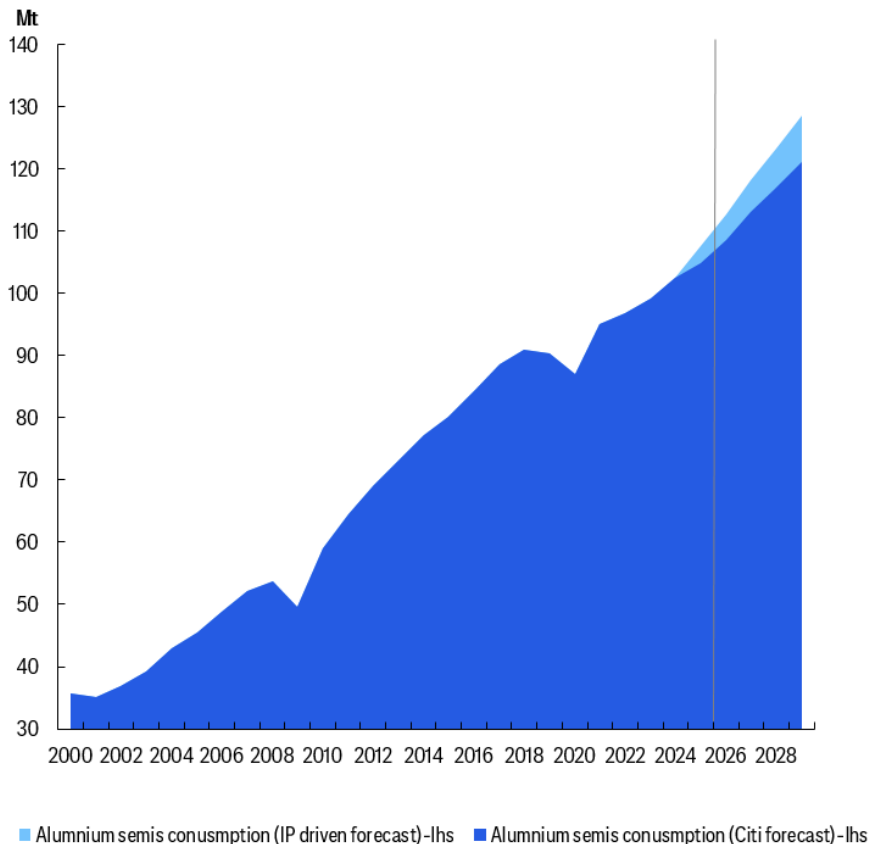
Our baseline consumption growth forecasts are relatively conservative

Our base case aluminium semis consumption growth is conservative compared with the IP-driven forecast based on historical trends. Considering our macro assumptions - namely, an eventual return of cyclical demand growth alongside continued decarbonisation-driven demand growth - we believe our forecast remains reasonable. Historically, even in periods such as the 2010s when decarbonisation demand was minimal, overall demand could still grow by >3Mt/y. Given the additional contribution from decarbonisation-driven growth today, we think our conservative forecast is well supported.

Our aluminium demand growth forecasts are conservative relative to their historical relationship with industrial production growth

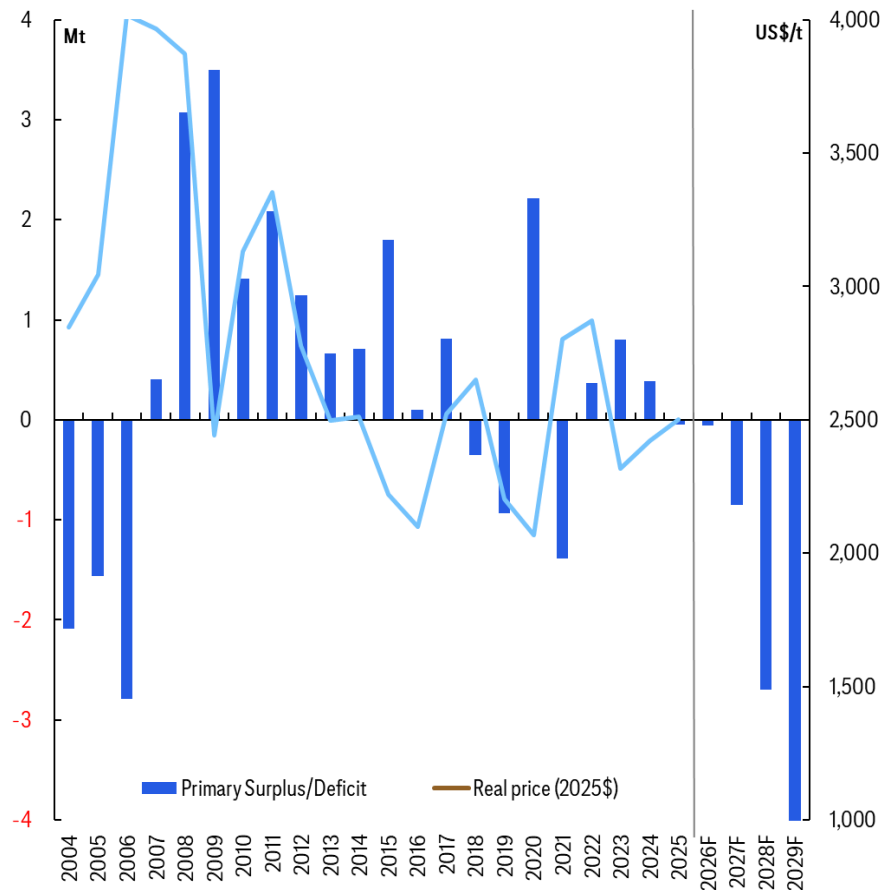


Our aluminium demand forecasts are conservative relative to industrial production



Aluminium is heading for a stock-out should prices remain at current levels

Primary aluminium supply and demand balance
(i.e., first iteration - pre-supply/demand response to higher prices)



Key Supply and Demand line items and Balances point to massive deficits by 2027 (i.e., first iteration - pre-supply/demand response to higher prices)

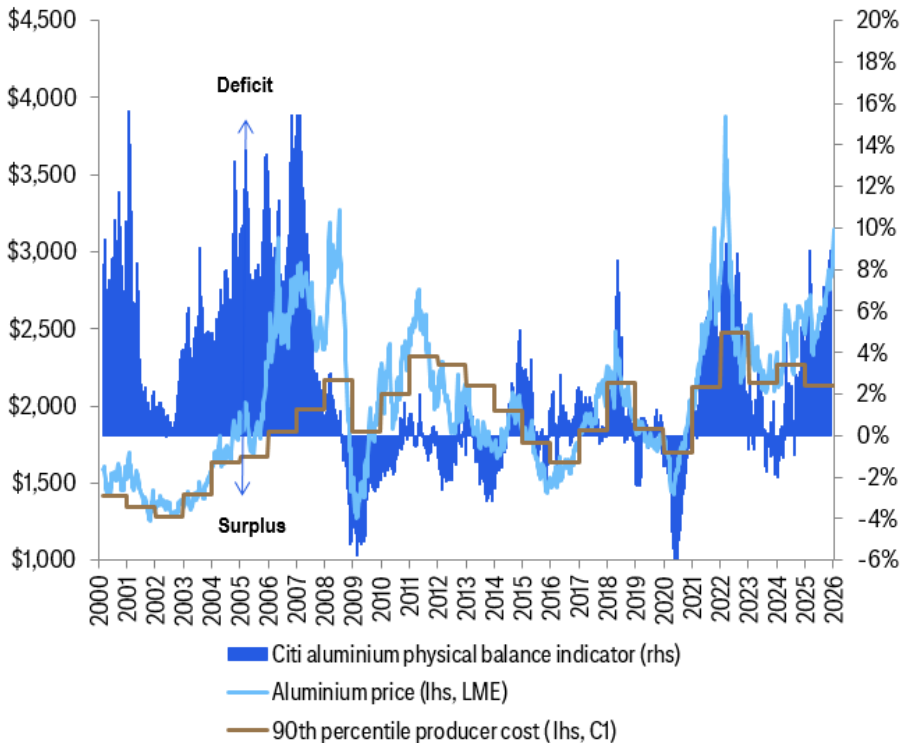
Aluminium Supply and Demand Balance (kt)

kt	2018	2019	2020	2021	2022	2023	2024	2025	2026F	2027F	2028F	2029F
Total Aluminium Consumption	90972	90362	87136	95003	96937	99121	102479	105404	108970	113054	116678	120361
y/y chg	2.8%	-0.7%	-3.6%	9.0%	2.0%	2.3%	3.4%	2.9%	3.4%	3.7%	3.2%	3.2%
Recycled Metal Consumption	27,989	28,053	26,210	27,923	30,430	31,240	31,882	33,541	34,881	37,145	39,116	40,885
y/y chg	2.8%	0.2%	-6.6%	6.5%	9.0%	2.7%	2.1%	5.2%	4.0%	6.5%	5.3%	4.5%
Primary Consumption	65,024	64,416	62,618	68,816	68,577	69,953	72,516	73,975	76,202	78,050	79,737	81,821
Consumption Incr. (%)	3.0%	-0.9%	-2.8%	9.9%	-0.3%	2.0%	3.7%	2.0%	3.0%	2.4%	2.2%	2.6%
Of Which China	35,692	36,018	37,466	40,047	40,183	42,684	44,710	45,934	47,051	47,929	48,725	49,890
Consumption Incr (%)	3.9%	0.9%	4.0%	6.9%	0.3%	6.2%	4.7%	2.7%	2.4%	1.9%	1.7%	2.4%
Primary Production	63,998	63,286	64,751	67,402	68,843	70,689	72,774	73,929	76,141	77,203	77,037	77,498
Supply Incr (%)	0.5%	-1.1%	2.3%	4.1%	2.1%	2.7%	2.9%	1.6%	3.0%	1.4%	-0.2%	0.6%
Of Which China	36,317	35,352	36,745	38,524	40,085	41,584	43,069	43,777	45,067	44,957	44,470	44,488
Supply Incr (%)	-0.3%	-2.7%	3.9%	4.8%	4.1%	3.7%	3.6%	1.6%	2.9%	-0.2%	-1.1%	0.0%
Of Which Indonesia	242	250	245	243	229	372	533	805	1,206	1,527	1,570	1,570
Supply Incr (%)	11%	3%	-2%	-1%	-6%	63%	43%	51%	50%	27%	3%	0%
Smelter Capacity (kt/y)	75,906	73,781	75,132	76,237	78,553	79,153	79,373	80,086	81,780	82,179	82,662	83,257
Primary Surplus/Deficit	-356	-933	2216	-1384	366	798	385	-46	-61	-847	-2700	-4323
China	643	-705	339	81	392	313	396	338	732	32	-1,212	-2,355
Ex-China	-999	-227	1,877	-1,465	-26	485	-11	-384	-792	-879	-1,488	-1,968
US\$/t	2,114	1,795	1,705	2,476	2,707	2,255	2,425	2,620	3,025	3,500	3,800	3,200

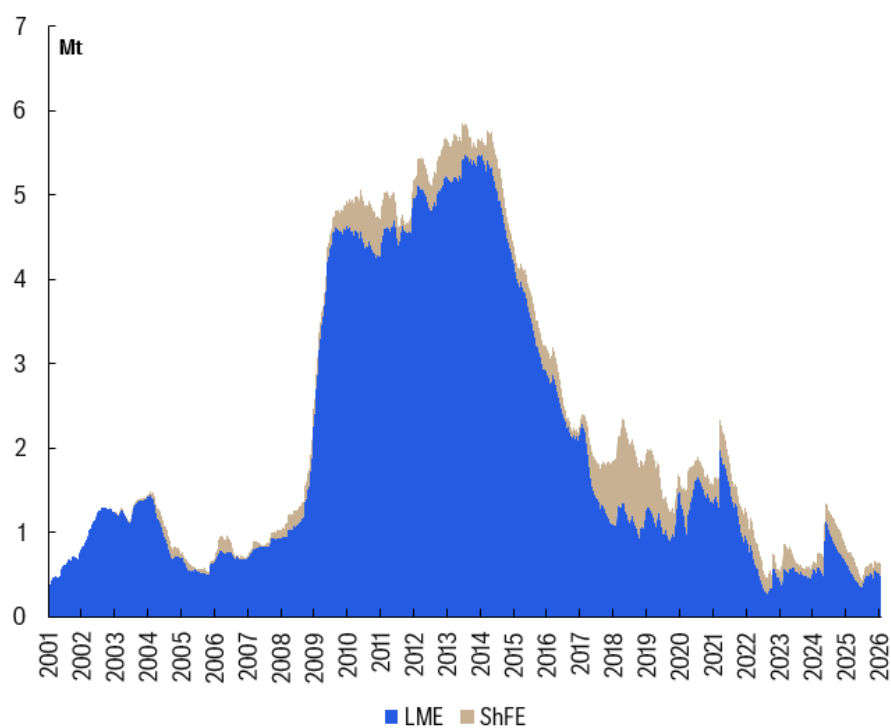
Stocks and convenience yields point to market tightness, but not convinced yet

Our physical balance indicators, derived from convenience yields, shows the market repeatedly flipping into deficit when inventories tighten, however this may reflect net producer hedging activity and we are not getting overly excited about this indicator. Having said this, reported stocks across major markets have collapsed to multi-year lows, leaving little buffer to absorb future deficits or any further supply shocks.

Aluminium physical balance indicator tracks scarcity via convenience yields



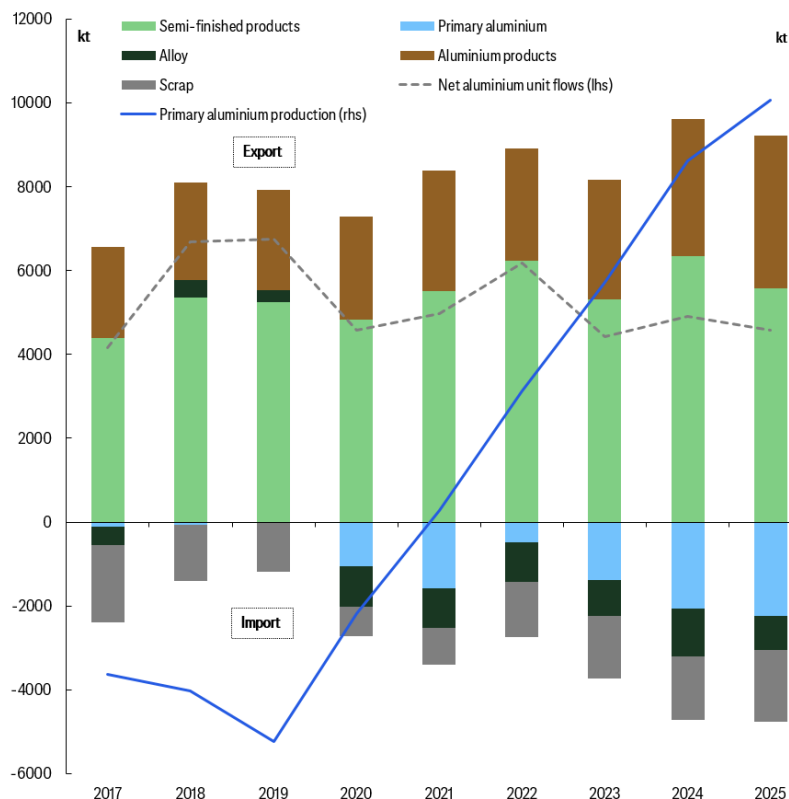
Reportable inventories have decreased to multi-year lows



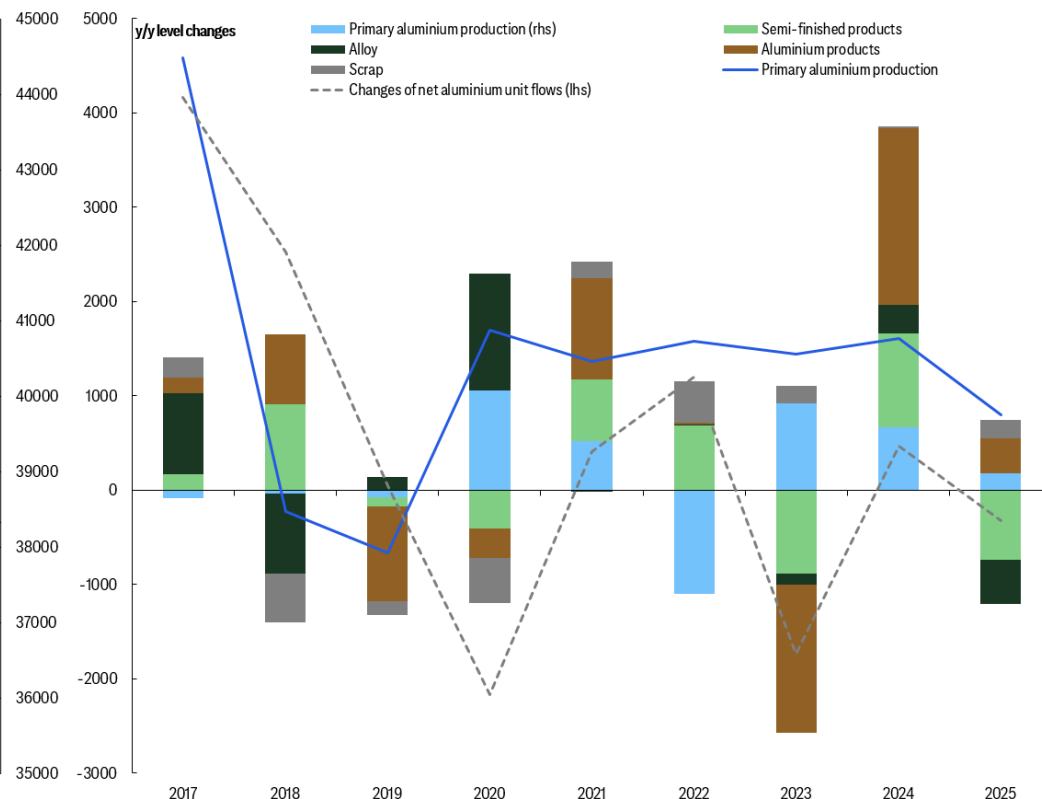
Ex-China balances appear set to tighten as China's net exports decline

China's primary aluminium production growth is slowing sharply, and we project it to reach zero by 2027, indicating that the country's domestic supply elasticity has been significantly constrained. In contrast, China's exports flexibility has remained substantial in recent years - not only through direct exports of semi-finished products and fabricated products, but also via a much larger share embedded in a wide range of manufactured goods. This structural imbalance suggests that China may increasingly need to rely on imports of primary metal/alloy/scrap to cover potential shortfalls once domestic growth stalls.

Post-2020 export expansion intensifies China's demand for aluminium units, calling for more imports



China's primary aluminium production growth plateaus as export volatility persists



Aluminium supply – constrained by
competition for power, China capacity cap

Aluminium costs of production under AI/datacentre power pricing \$~3.5k/t!

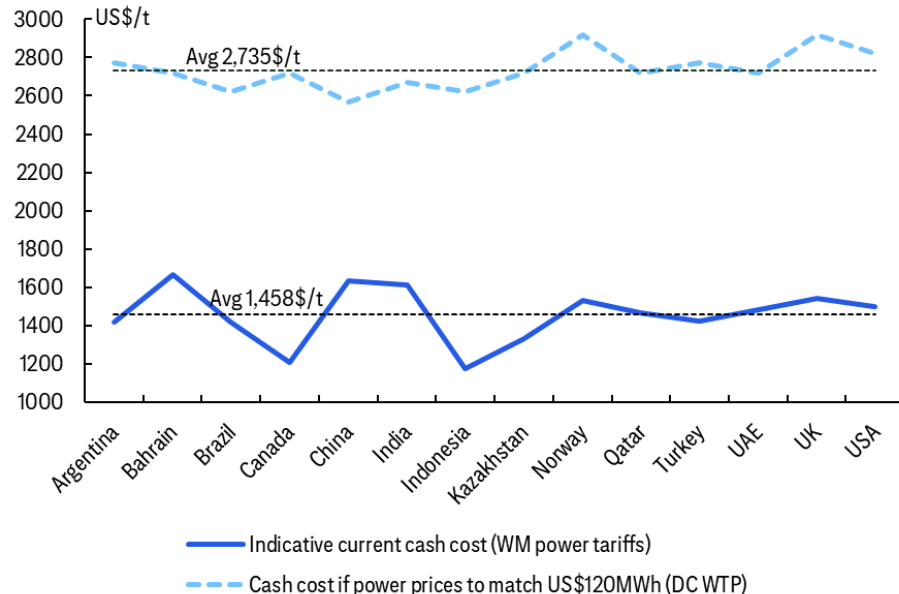
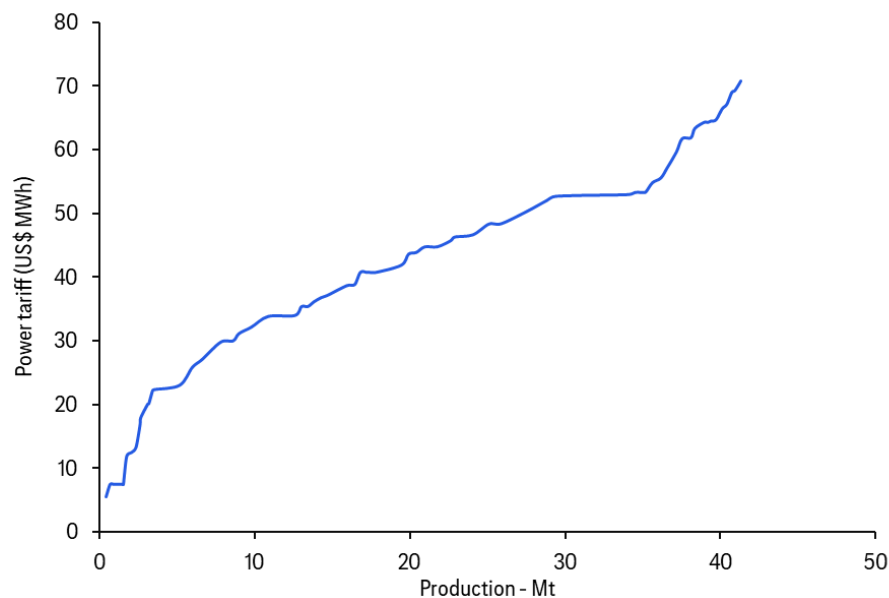
The growing overlap between aluminium smelting and hyperscale data centres offers an unconventional lens for assessing smelter economics. Both are highly electricity-intensive, and DC-style willingness-to-pay (WTP) levels are now materially above legacy industrial tariffs. Repricing smelter power to DC WTP levels therefore provides an indication of the potential scale of uplift embedded in today's development valuations.

A stylised 'cash-cost repricing' exercise shows that replacing Wood Mackenzie's smelter power tariffs with a uniform US\$120/MWh DC-WTP benchmark would lift implied cash costs to the US\$2,600–2,950/t and total costs to \$3,300–3,500/t across nearly all regions. This is not a forecast but illustrates the distance between current cost structures and DC-style valuations.

The uplift does not translate directly into smelter stress because most smelters operate under long-dated PPAs, captive-power systems, or regulated frameworks that insulate them from market-driven power prices. Only merchant-exposed regions—primarily parts of the US and Western Europe—sit close to the interface where tightening power markets could meaningfully reshape aluminium economics.

Most global aluminium smelters sit near ~US\$40/MWh (weighted avg); tariffs reflect legacy PPA/captive supply, not market power prices

Repricing smelters power to US\$120/MWh would sharply lift cash costs across all regions

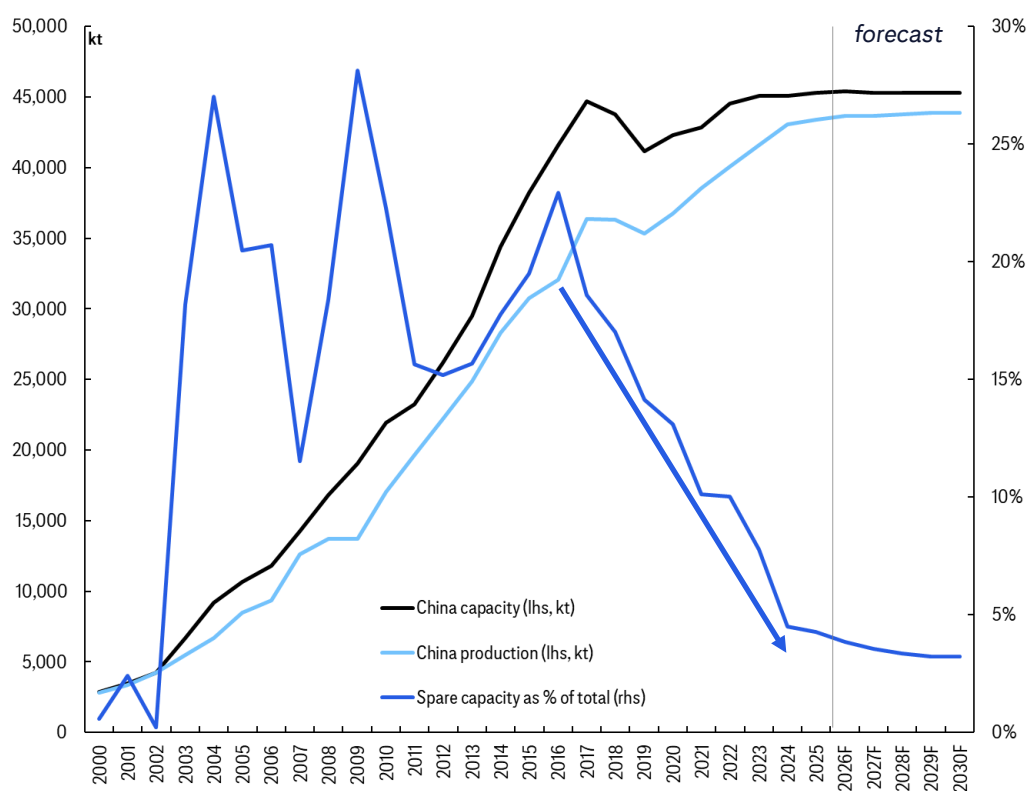


China is now likely ex-aluminium supply growth

From the early 2000s through to 2017, the aluminium market was flooded with low-cost Chinese capacity, driving prices and margins sharply lower. It has taken almost a decade, but this historical spare capacity has now been fully absorbed. As a result, China is no longer the marginal source of supply growth.

Initially, the slowdown in capacity additions reflected weak margins (2017–20). Since the early 2020s, however, the lack of new additions has been driven primarily by policy constraints — not only the 45 Mt capacity cap, but also environmental, energy and carbon policies. Together, these have made incremental aluminium smelting capacity in China a structurally poor investment, in our view.

Historical spare capacity in China's aluminium industry has disappeared - implications for price and supply elasticity

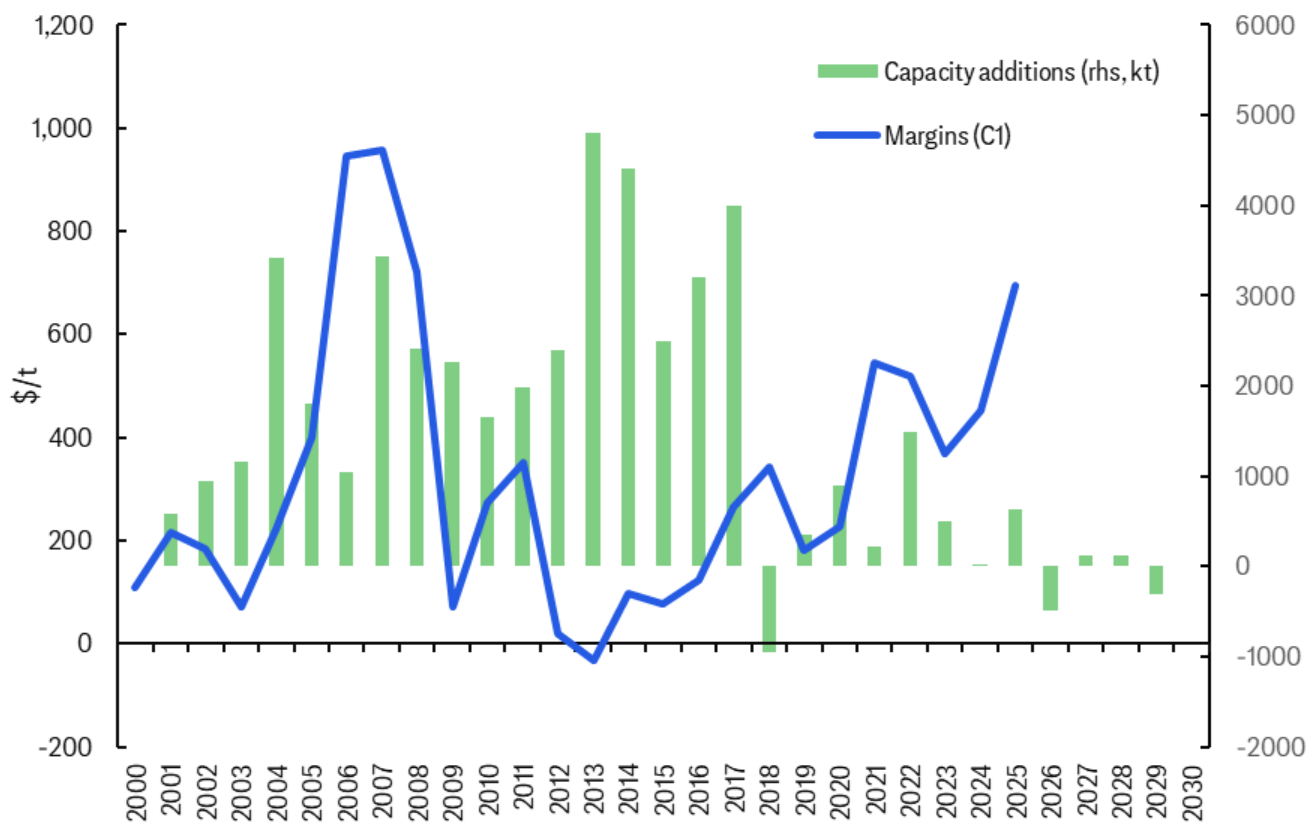


China is very unlikely to raise its capacity cap, meaningful cheating is unlikely

Clear policy signals China's capacity cap, and a very high bar for breaking it. Our base case is that China will maintain its de-facto aluminium capacity cap by not approving any new or additional projects, and we believe there is a very high bar for breaking it. After clearing illegal projects (incl capacity swapping) during 2013-2017, legal capacity reached 45.43Mt, forming a de facto capacity ceiling. Breaking it would require extraordinary pressure (e.g. acute shortage, soaring prices, heavy dependence on imports).

Conditional and trade-off increase in output through increased amperage. Existing installed capacity could theoretically rise ~10% by increasing amperage, but constrained by cell design, power supply, and environmental limits, while incurring higher costs, faster equipment wear, and potential safety issue.

China's low-cost expansion era ends, driving high China utilisation and strong margins



Indonesia supply growth is expected to be ~0.7mtpa, vs ~2-3mtpa required

There are ambitious plans to build primary aluminium smelters in Indonesia, driven by both push factors (China's capacity cap) and pull factors (Indonesia's ban on bauxite exports). Chinese producers also view this as an important step in diversifying supply risk, given China's heavy reliance on bauxite imports from Guinea.

However, the pace of Indonesia's expansion is not only a function of economics, but is heavily dependent on execution and timing, in particular the ability to develop captive power infrastructure and bring associated smelting capacity online in parallel. Access to affordable and reliable power—primarily via captive coal-fired generation—remains a critical gating factor, alongside regulatory, ESG and financing considerations.

Indonesia aluminium smelter projects-development tracker

Project / Smelter	Location	Type	Power Source	Capacity +/- within a year					Targeted Capacity (ktpa)	First Production
				2025e	2026f	2027f	2028f	2029+f		
Asahan/Kuala Tanjung	North Sumatra (Kuala Tanjung)	Existing project	Hydro							1984
	North Sumatra	Expansion	Coal	Base				100	300	2029+
Kalimantan Aluminium	North Kalimantan	Greenfield	Coal	Poss					500	2030+
Alamtri	North Kalimantan	Greenfield	Coal	Base	250	250			1500	2025+
Harita/DIB Aluminium	West Kalimantan	Greenfield	Coal	Prob		100			1000	2027+
NA	West Kalimantan	Greenfield	Coal	Poss					2400	2030+
Huaqing	Central Sulawesi (IMIP)	Existing project	Coal						500	2023
Huaqing	Central Sulawesi (IWIP)	Expansion	Coal	Poss					500	2030+
Taijing	Central Sulawesi (IMIP)	Greenfield	Coal	Base	300	300			600	2026+
Juwan	North Maluku (IWIP)	Greenfield	Coal	Base	280				1500	2025+
Xianfeng	North Maluku (IWIP)	Greenfield	Coal	Base	320					2026+
	North Maluku (IWIP)	Expansion	Coal	Poss			250	500		
Bintan	Bintan Island	Greenfield	Coal	Base		50	200	250	1000	2027+
Huayou	NA	Greenfield	Coal	Poss					500	2030+
New capacity additions (base)				280	870	600	200	350		
Installed capacity (base)				1080	1950	2550	2750	3100		
Output (base)				805	1530	2147	2190	2190		
New capacity additions (base+prob+poss)				280	870	700	450	850		
Installed capacity (base+prob+poss)				1080	1950	2650	3100	3950		

Even a full Indonesia ramp-up would not close the global supply gap

With China capped at 45 Mt, Indonesian projects will play a critical role in alleviating supply tightness. However, even full project realisation would only partially narrow the gap, leaving the market exposed to potential deficits through 2030. This outlook also assumes faster growth in secondary aluminium supply and demand, helping offset part of the requirement for primary metal to meet underlying consumption growth.

We expect Indonesia's aluminium capacity additions to support steady production growth of around 0.5 Mt per year over the next three years. China's Tsingshan is a major investor in Indonesia's aluminium sector and, through joint ventures with partners, is expected to deliver significant new capacity, supported by its established industrial parks linked to nickel operations. However, we do not believe Indonesia can replicate the pace of capacity build-out China achieved in the early 2000s.

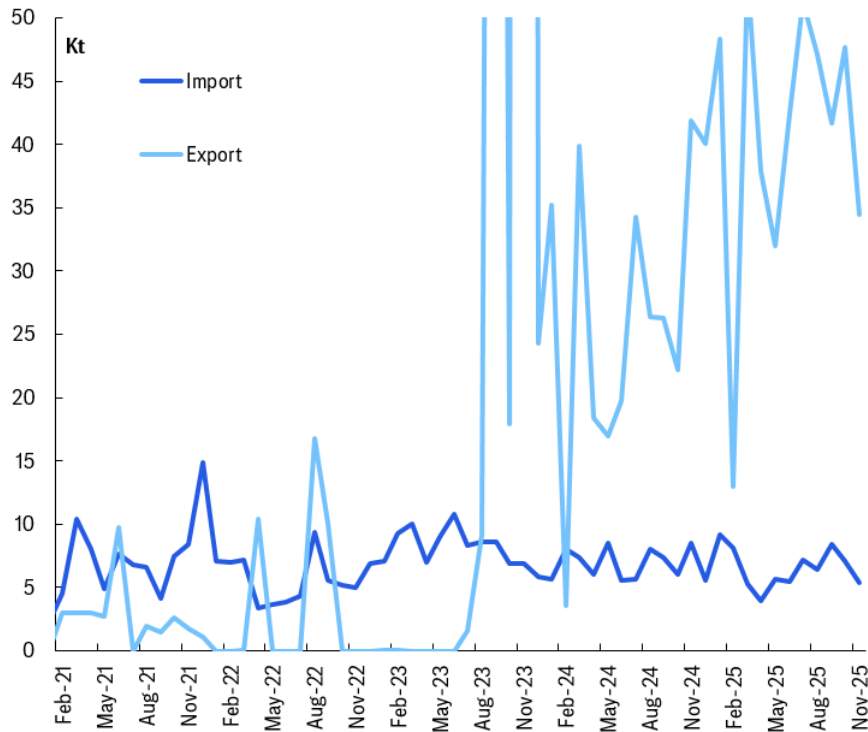


Most aluminium smelting projects in Indonesia are concentrated in resource-rich but relative underdeveloped provinces. Their industrialisation relies heavily on large-scale industrial parks, dedicated power supply, and port infrastructure. These projects act as catalysts for regional development rather than being located in already industrialised areas. The notable exception is Bintan in the Riau Island, which benefits from proximity to Singapore and a well-established free trade zone, giving it a stronger economic foundation compared to other projects sites.

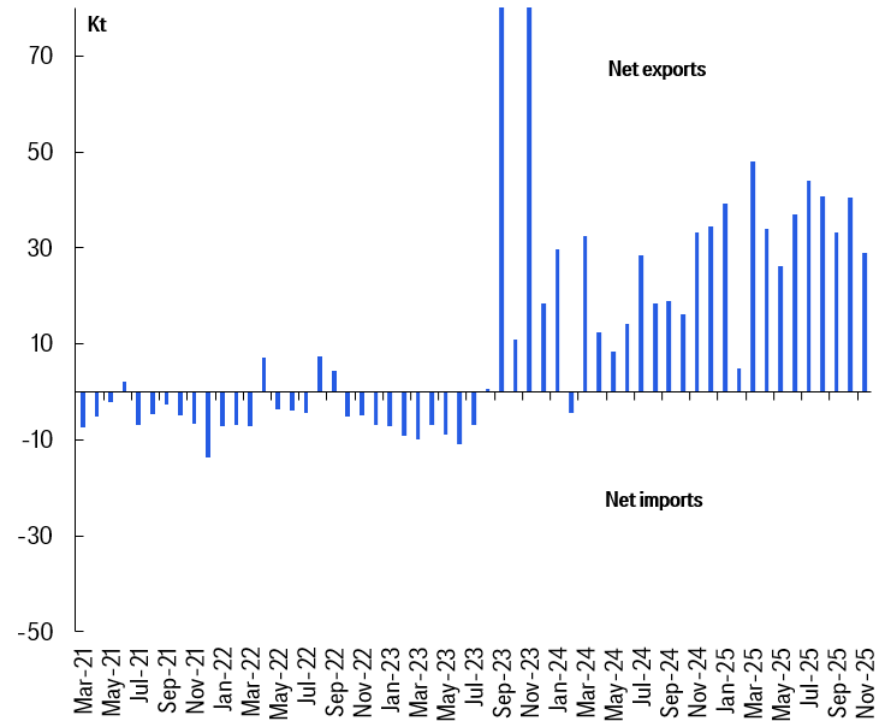
Indonesia emerges as a marginal net exporter

Indonesia's capacity additions, particularly the 2023 commissioning of the Huafon project, has placed the country firmly in a structural net-export position, with year-to-date net exports annualised at just above 400kty and China remaining the dominant destination. Tracking Indonesia's export flows will be increasingly important for accessing incremental supply to the global market, as upcoming projects are expected to rely primarily on China as their main offtake.

Indonesia unwrought aluminium imports and exports



Indonesia's monthly net export of unwrought aluminium

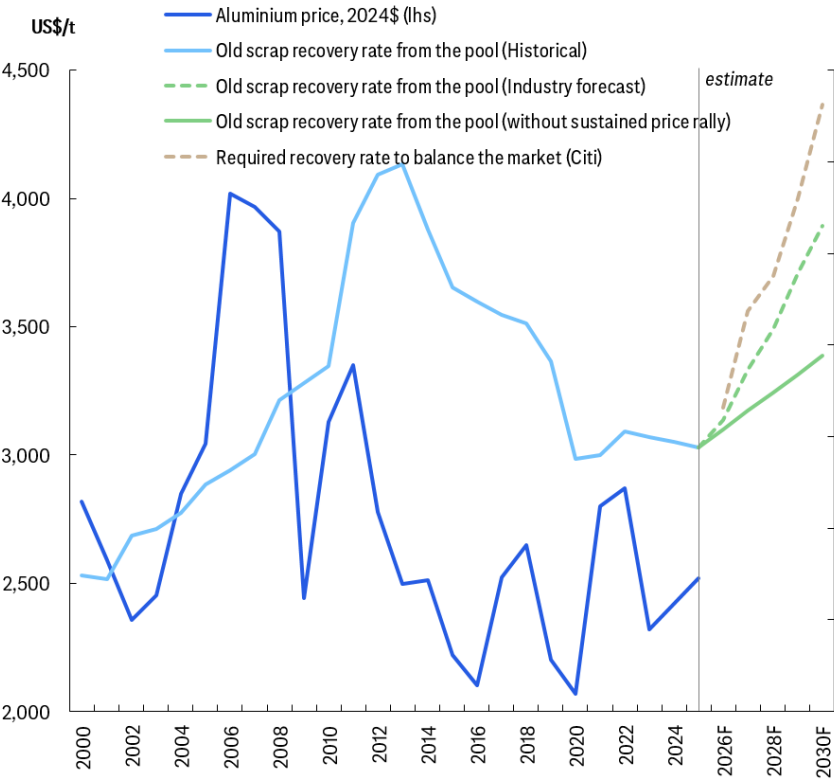


Secondary supply will grow, but insufficient at current prices

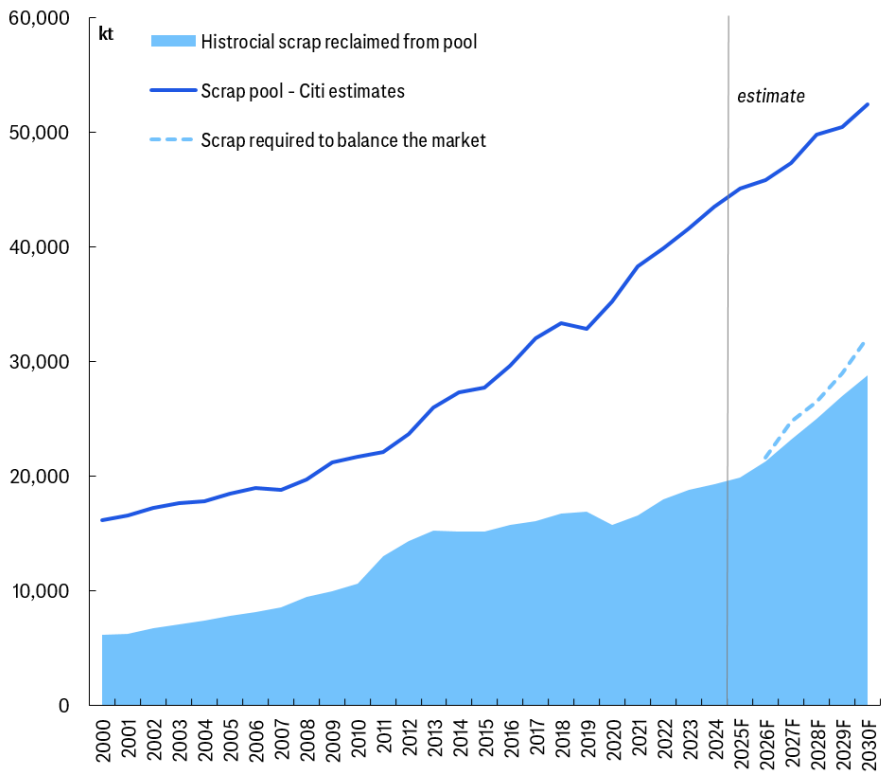
The surge in LME aluminium prices in early 2000s drove a higher share of old scrap recovery, albeit with some lag. Since peaking in 2013, however, the recovery rate has declined in the absence of strong price rally and has hovered around ~45% over the past five years. This highlights a critical issue: if the market continues to heavily rely on secondary supply, further growth in recycling rates will likely require similarly strong prices incentives – a challenge that we think is easier said than done.

Without a sustained price rally, we expect only a moderate increase in recovery rate (~50%) in the coming years, reflecting efficiency gains from AI-based sorting technologies. Still for the market to achieve balance, we believe the recovery rate will need to rise above the previous cycle's peak.

History suggests no breakthrough in scrap recovery without strong price signals



With a flat recovery rate, scrap reclamation from the pool will be insufficient to meet market balance requirement



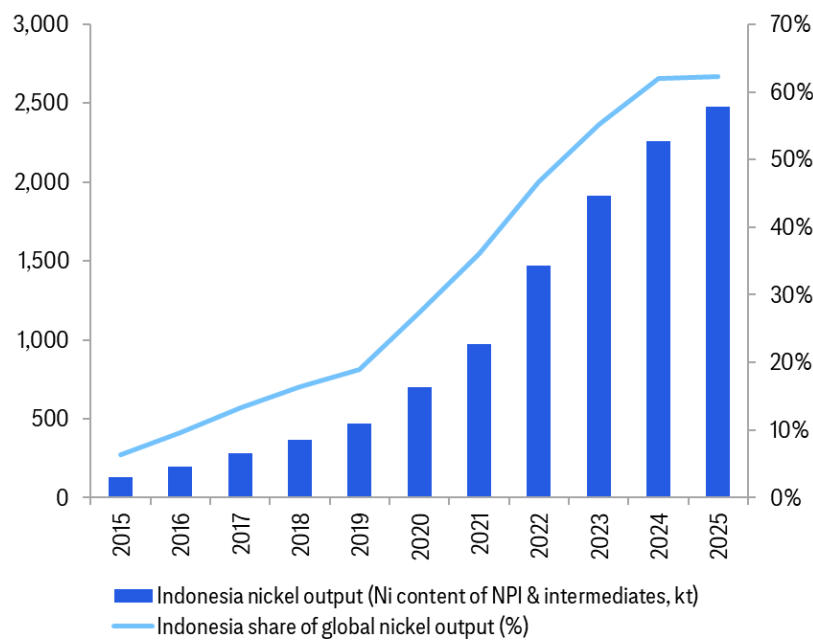
If Indonesia nickel build out is replicated supply grows only ~0.7mtpa!

We do not see Indonesia supply growth dominating the global aluminium market like it has done for nickel over the last 10 years. To do so would require development of 5x the power capacity associated with the domestic nickel boom in just 5 years. If Indonesia's aluminium supply growth was constrained by power growth equivalent to nickel's decade-long expansion it translates to ~7Mt in 10 years (~0.7 Mtpa). Even double that pace (~1.4 Mtpa) would fall short of trend total aluminium demand growth of ~3Mtpa.

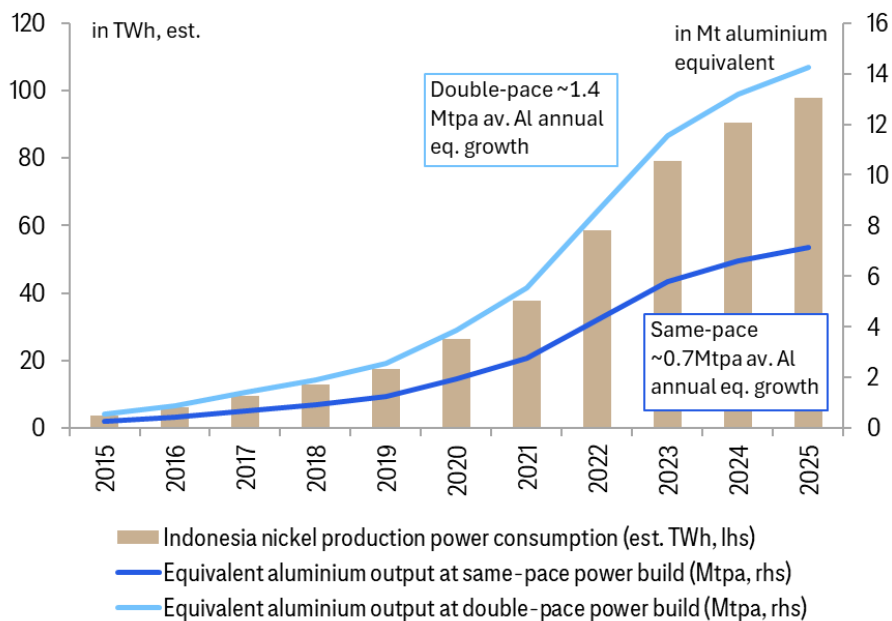
Indonesian aluminium output expansion could ramp much more quickly in the very unlikely scenario that current installed power for domestic nickel industry was redirected towards aluminium – this would be net bearish aluminium balances and price, bullish nickel, but this is very unlikely to occur on any meaningful scale. Despite depressed nickel prices and cash-negative NPI margins, a value-chain-wide perspective suggests no overall incentive to unwind Indonesia's nickel system at scale.

However, in the medium-to-long term, competition for capital, power, and labour with a burgeoning domestic aluminium industry could support some degree of constraint and cost floor on a future incremental build-out of Indonesia's nickel sector. [Metal Matters – Is nickel a proxy or avenue for Indonesian aluminium supply? Plus aluminium costs in a datacentre power price scenario](#)

Indonesia took ten years to lift nickel from <10% to ~60% of global supply; but it now dominates in a surplus global market



If an Indonesian aluminium capacity build-out matched nickel in power terms, it would equate to ~0.7 Mtpa, or 7Mtpa over a decade.

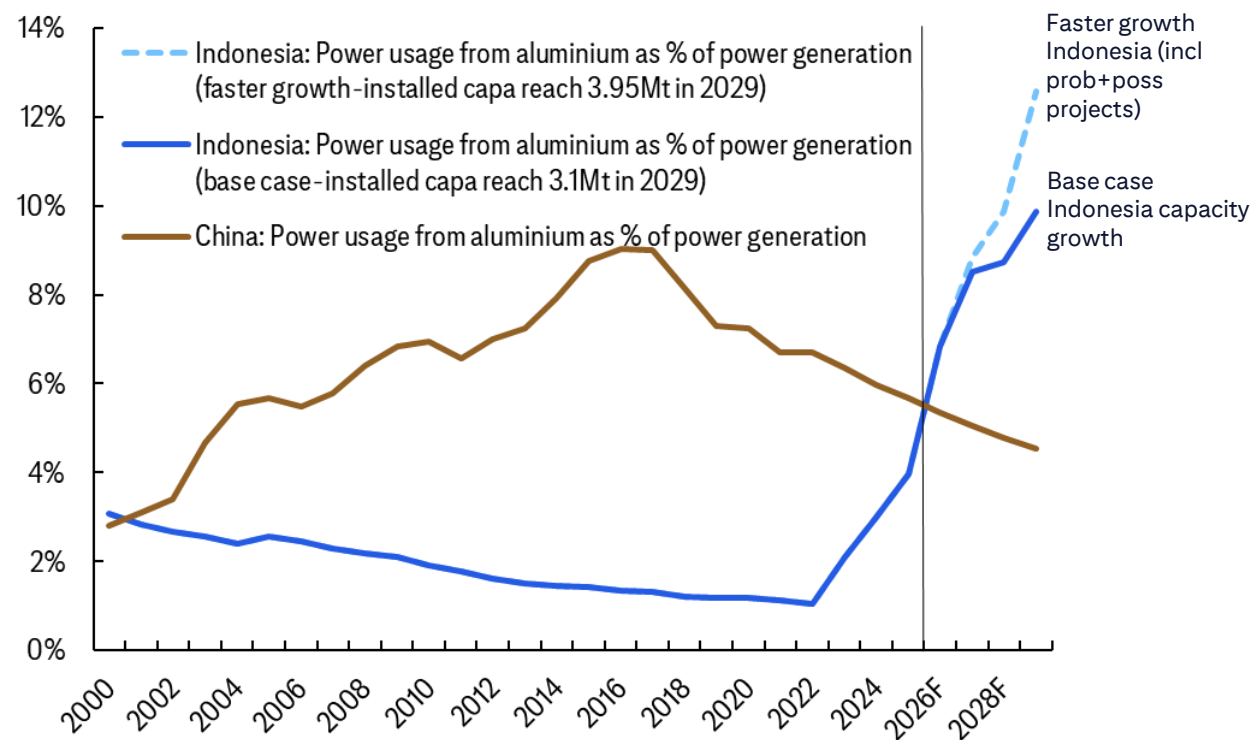


Indonesia's ambitious aluminium plans - a power reality check

Based on the latest RUPTL and national power-generation forecasts, our base-case outlook suggests aluminium installed capacity could reach c.3.1Mtpa by 2029. At this level, the sector would consume around 10% of national power generation. While this is comparable to China's 2016–17 peak of ~9%, it would place substantial pressure on Indonesia's base-load system, particularly given potential overlaps between reported grid availability and captive supply.

This implies that almost all incremental aluminium capacity will need to rely on captive power build-out rather than the grid. However, captive power development typically requires long lead times, particularly where financing conditions are tight and equipment delivery cycles are extended. Against this backdrop, project delays would not be surprising.

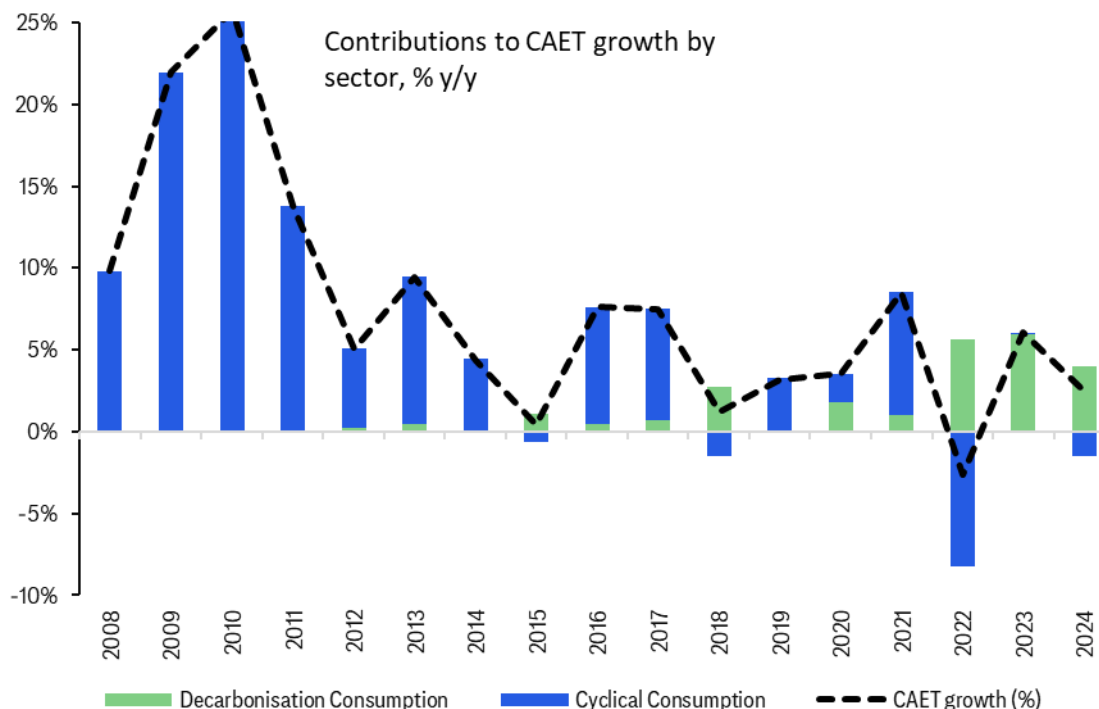
Indonesia's aluminium power usage (as % of power generation) - Base case manageable, fast growth scenario would push beyond China's peak levels



Aluminium – Decarb demand growth offsetting cyclical weakness

	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25
China Aluminium End-Use Tracker (CAET) - % change y/y	3.1%	5.7%	4.4%	10.1%	4.1%	8.5%	4.4%	6.1%	10.1%	4.3%	2.3%	3.4%	2.8%
China Aluminium End-Use Tracker (CAET) - % change ytd	1.1%	1.5%	1.8%	2.5%	4.1%	6.3%	5.7%	5.8%	6.6%	6.2%	5.7%	5.4%	5.1%
Transportation	1.5%	12.8%	17.4%	20.9%	2.5%	37.9%	12.1%	16.3%	14.0%	18.6%	20.2%	19.6%	22.2%
Construction	-7.2%	-7.1%	-12.3%	1.5%	4.9%	-5.3%	-2.6%	-11.2%	-10.9%	-6.7%	-18.3%	-18.2%	-18.2%
Packaging	5.6%	5.0%	6.5%	10.7%	-3.6%	-3.6%	-2.4%	0.8%	-2.5%	-2.7%	2.2%	-3.4%	-2.8%
Foil stock	9.1%	9.1%	9.1%	9.1%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%
Electrical	14.5%	17.3%	18.0%	20.5%	12.6%	25.6%	11.6%	49.9%	97.4%	-1.2%	-3.5%	4.0%	0.5%
Consumer durables	10.8%	9.3%	8.1%	16.0%	8.4%	8.4%	4.1%	-2.2%	0.5%	5.7%	6.7%	6.2%	-0.7%
Machinery & Equipment	3.2%	6.1%	1.9%	8.2%	-4.1%	-4.1%	1.1%	-1.5%	-0.6%	2.9%	3.2%	7.9%	3.1%
Other	5.4%	5.3%	5.4%	6.2%	5.9%	5.9%	7.7%	6.1%	5.8%	6.8%	5.7%	5.2%	6.5%
Cyclical	-3.8%	-1.5%	-3.6%	3.2%	0.0%	2.8%	-1.7%	-2.3%	-0.5%	1.3%	-2.2%	-1.6%	-2.8%
Decarbonisation	32.1%	34.5%	35.3%	29.7%	31.8%	59.3%	31.7%	42.7%	50.1%	14.5%	18.5%	21.1%	19.9%

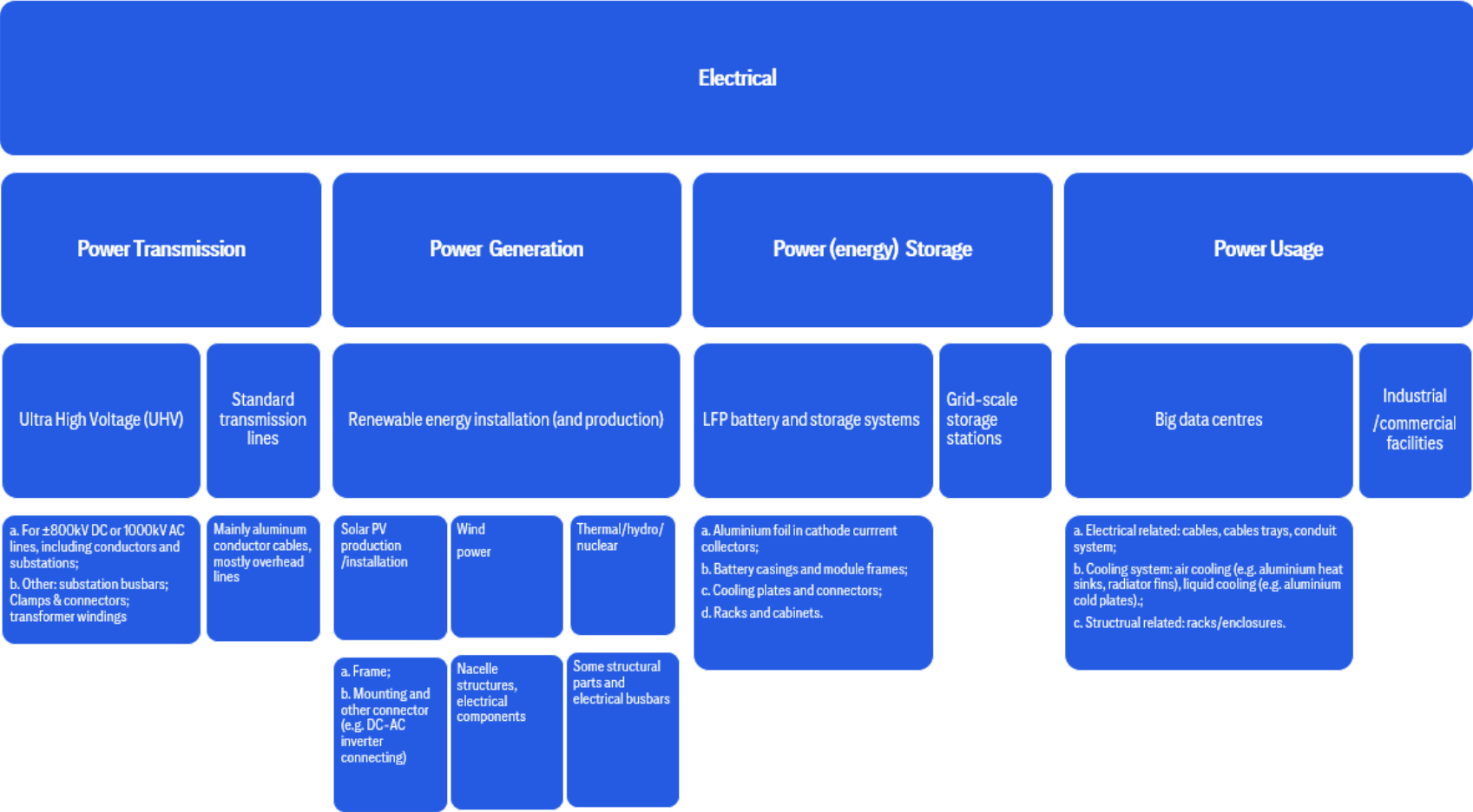
Contributions to global aluminium end use demand growth by segment (%)



Aluminium's structural demand drivers – power infrastructure, datacentres, robots, batteries, solar

China’s electrical sector – four pillars driving aluminium demand

China’s electrical sector has emerged as a key driver of aluminium demand growth, underpinned by strong policy momentum and evolving industrial priorities. We break this sector into four core pillars: power transmission, power generation, power storage, and power usage. Together, these segments form the backbone of China’s energy transformation. The sector’s robust expansion is primarily fuelled by the country’s decarbonisation agenda and President Xi’s strategic push for a “new productive force,” which includes building out AI-driven data centres. These facilities require massive power consumption and, in turn, significant investment in electrical infrastructure — further reinforcing aluminium’s critical role in enabling the transition.

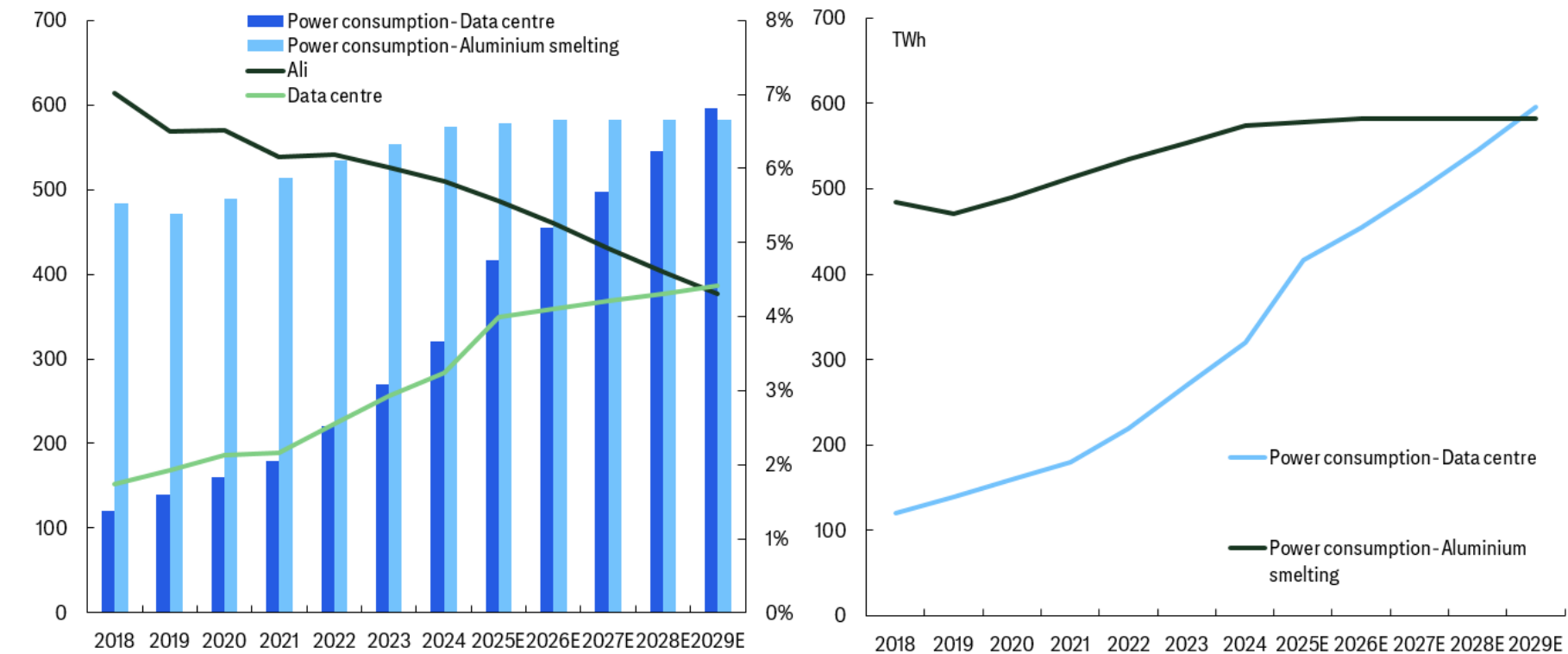


Data centers gain power priority over aluminium and is carbon restricted

Chin’s data centers are radially increasing their power consumption, a trend that is likely to accelerate further with the rise of AI-driven workloads (e.g., DeepSeek). This shift is transforming data centers from marginal users to major consumers—placing significant strain on the power grid and driving the need to expanded based-load capacity. This raises a critical question of which industry gets priority access? Data centers are more aligned with China’s strategic priorities, including digital infrastructure expansion, AI leadership and technological self-sufficiency. As a result, they are likely to enjoy fast-track access to power.

In contrast, aluminium smelting faces stricter energy controls under carbon reduction polices. The government is likely to maintain a hard cap on aluminium smelting capacity growth (i.e., 45Mt) despite rising demand. The latest campaign on anti-involution also reaffirms that China is unlikely to unwind its hard-won progress in this space. In addition, data centers are also facing mounting pressure to decarbonize, likely resulting in more aluminium demand.

Both aluminium smelting and data centre are hungry for baseload power supply



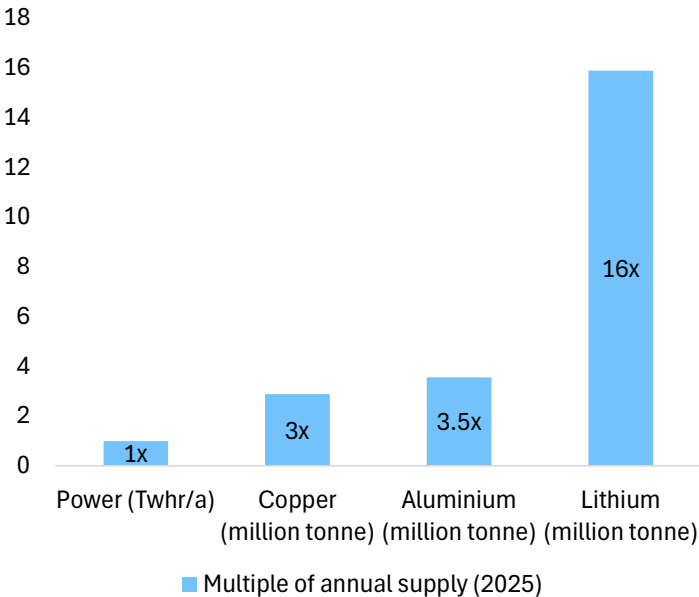
Robots – structurally bullish for power, lithium, copper, aluminium

Humanoid and non-humanoid robots are set to drive massive demand for power, lithium, copper, and aluminium over the medium to long term. Robot architecture is made from aluminium and other composites, and their power comes from lithium batteries, containing aluminium, recharged via copper cables. Robot intelligence requires power and metals intensive datacentres.

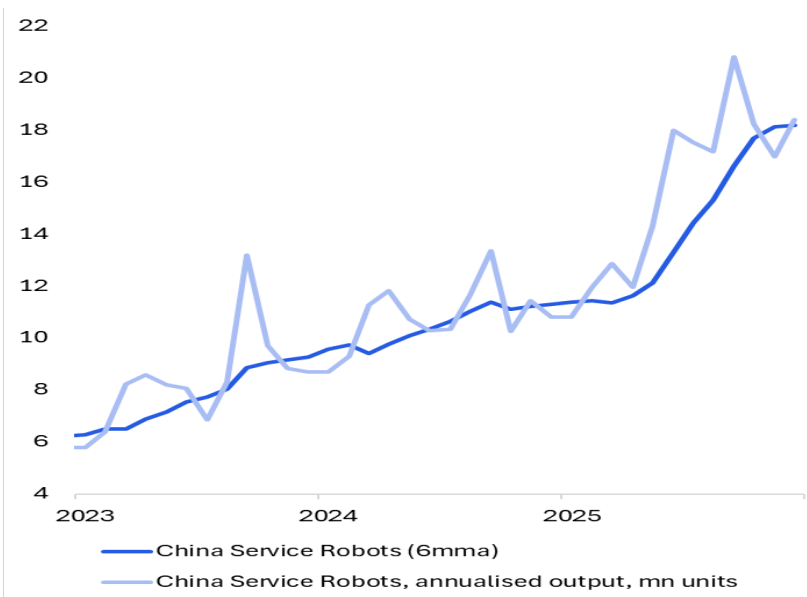
To put the potential scale of power and metals demand into perspective, we estimate that 10bn humanoid plus 10bn non-humanoid robots ([projected by Elon Musk by 2040](#), Reuters, Oct. 29) would require a doubling in global power output to recharge and that the production of these robots may require 16x, 3.5x, 3x, of lithium, aluminium, & copper supply respectively. N.B. estimates assume current tech.

China has very recently shifted capital allocation to the service (humanoid and non-humanoid) robot space, via its anti-involution drive and direct support from President Xi, with robot production up ~80% y/y as a result. China service robot production is now running at 18-20m annualised, up from 10-11m at the beginning of the year (lower right hand side chart). Over the coming years we expect capital allocation and robot output to boom and costs to fall, similar to what occurred during China’s energy transition (which began July 2020) and the US shale revolution.

Power required for, and lithium, aluminium, and copper contained in 10bn humanoid and 10bn non-humanoid robots (Elon Musk projection by 2040)



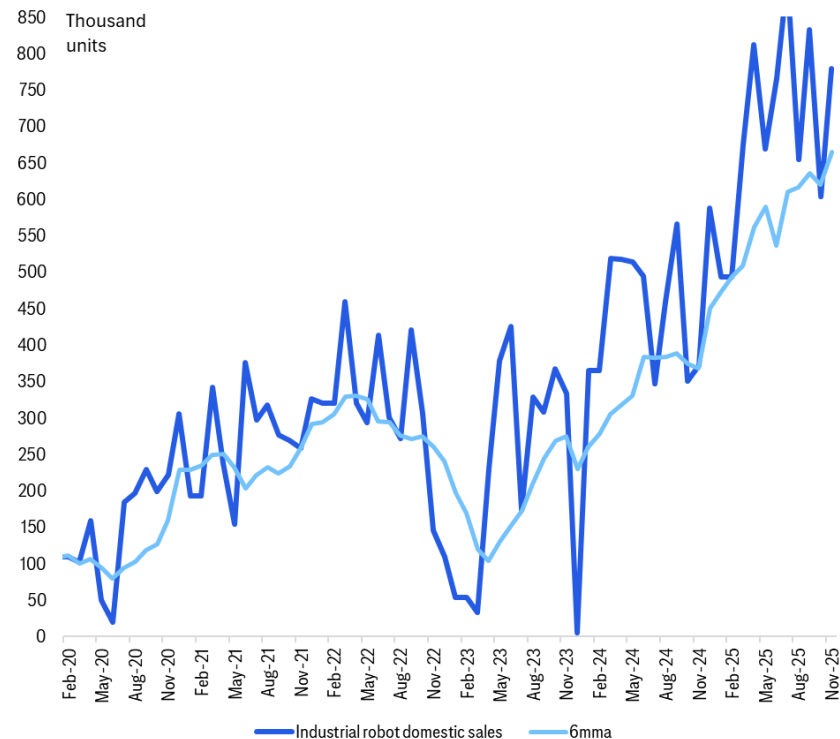
China’s service robot output has increased by 8 million units over the past year, with output accelerating sharply over the past 6 months



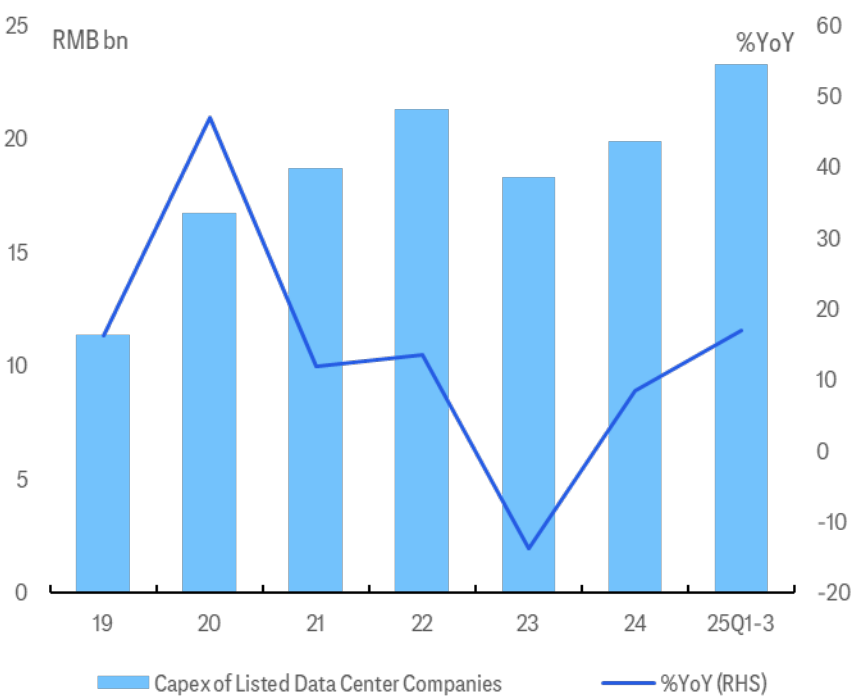
Robots, AI/data centres are booming in China, implications for AL, LI, power

China has dramatically stepped up its production of, and spending on, robots and data centres, a trend we expect to continue. Industrial robot output in China increased by ~60% y/y in July 2025 to 900k units (annualised), and data centre capex in China increased by 100% y/y in 1Q'25 (listed company capex). We believe these sectors should provide a medium- to long-term boost to aluminium, lithium, power, and copper demand. Capital is trapped in China and will be deployed in this space in a massive way, lowering costs, and driving technological innovation in these markets, providing a new growth engine alongside the energy transition spending that has helped to offset property market weakness. China's rare earth endowment may provide a competitive advantage in mass-scale robotics near term, in our view.

Industrial robot output in China increased by ~28% y/y in 2025 to ~8 units driven by strong domestic sales



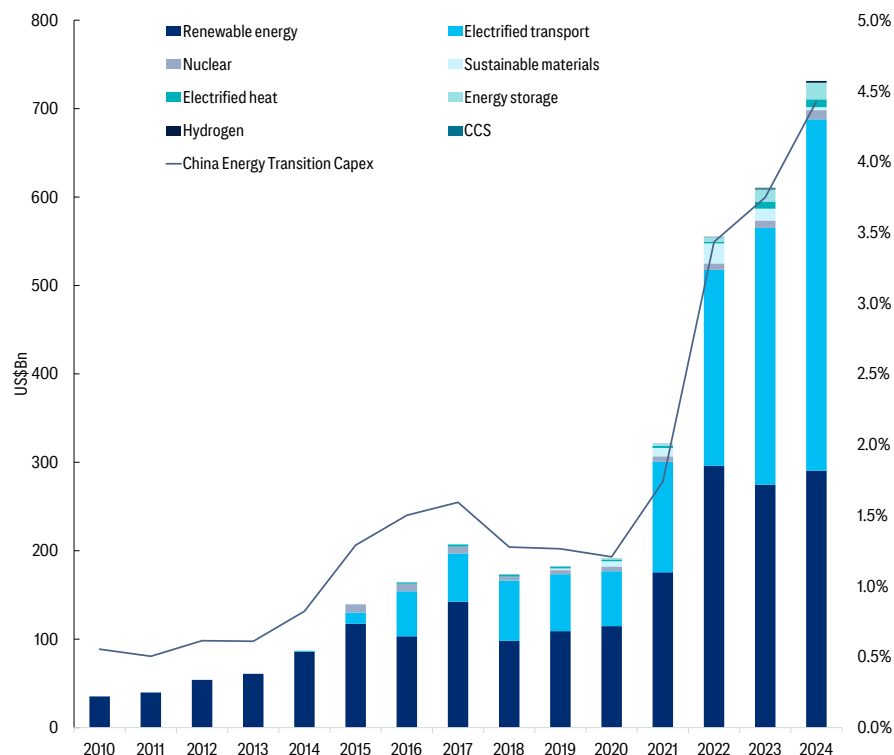
Data centre capex in China increased by 20% y/y in 1Q-3Q'25 (listed company capex)



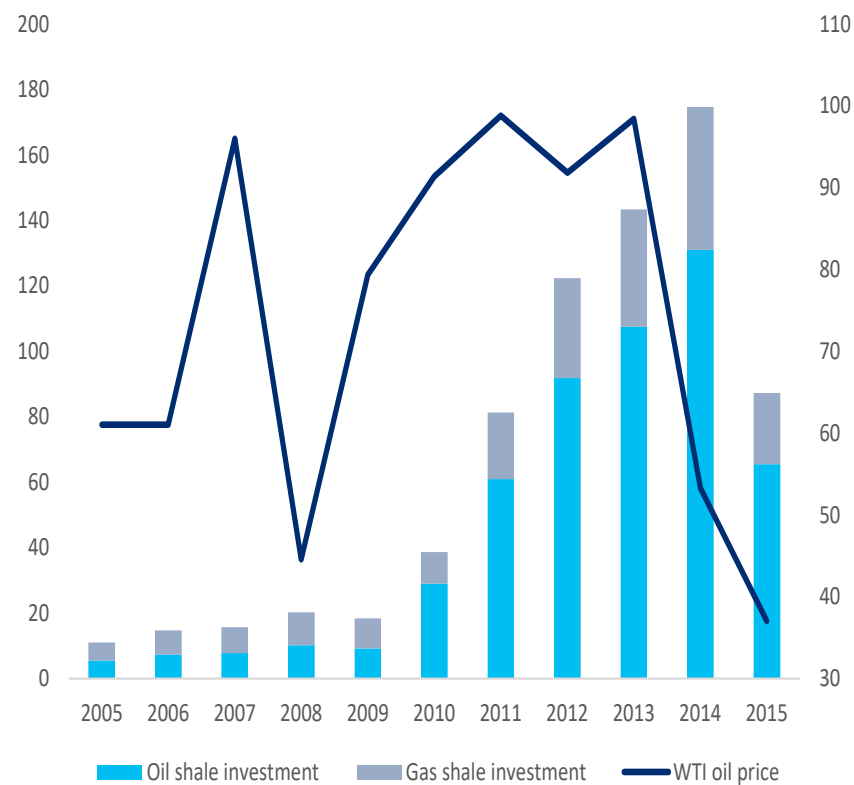
Robot/data centre boom may echo China's energy transition, shale revolution

When massive amounts of capital were directed at China's net zero (energy transition) push post July-2020 (left chart) and when massive amounts of capital were directed at US shale (right chart), we saw production boom and costs collapse (of electric vehicles, solar panels and installations, oil production). We could expect China to allocate massive amounts of capital to robotics (humanoid, dogs, aerial drones, industrial) and AI/data centres, and we could expect similar results. Commodity inputs for these sectors should boom, such as demand for aluminium, copper, lithium, and power.

Cumulative China investment in energy transition capex increased from less than 4-5x over the past 5 years to over \$700bn per annum



Cumulative US shale investment was ~\$500bn during the 2011-2014 period, driving the shale revolution (higher output, lower costs)

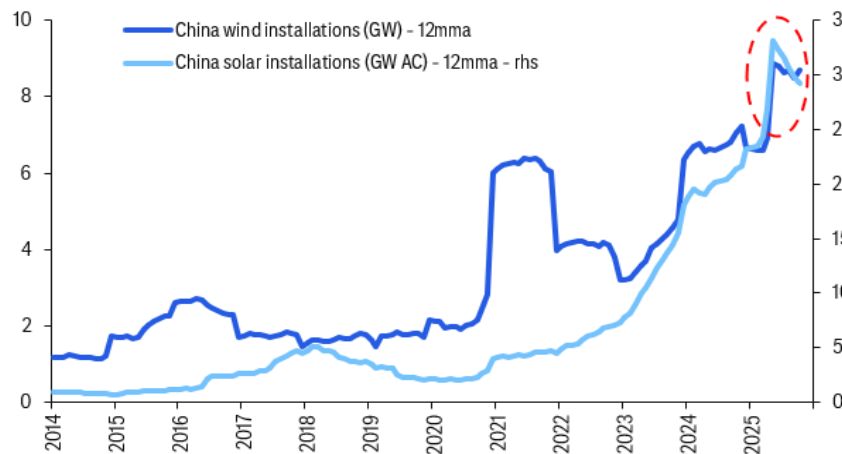


Global solar still growing despite recent fears

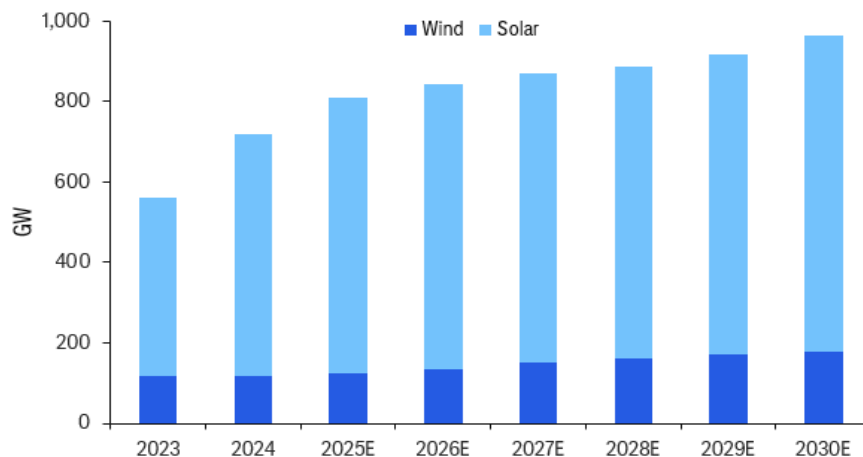
We see recent policy-induced softness in China solar installations since June as a blip and continue to project robust long-term global additions of solar and wind. Despite China policy changes as well as changes to US IRA clean tax credits, we expect healthy growth in the solar and wind installation pipeline over the medium-term. We expect renewable installation capacity additions (solar + wind) to grow from 800GWdc to ~1,000GWdc by 2030 in our base case, underpinned by strong growth in the electricity market.

China's reported solar and wind installations slowed through 2H'25 after a record 1H'25 surge following policy changes. The slowdown in renewable installations has impacted implied copper consumption growth as implied by our end-use trackers. Despite weakening from June, China's 10M'25 solar installations are still up 40% while wind installations grew >50%. We expect lower China domestic solar installations in 2026 to be more than offset by growth ex-China (with China metals consumptions still benefitting from related export demand).

China's solar and wind installations continue to surge despite the recent weakness



Citi commodities solar and wind capacity addition forecasts*
(Solar is GW/DC)

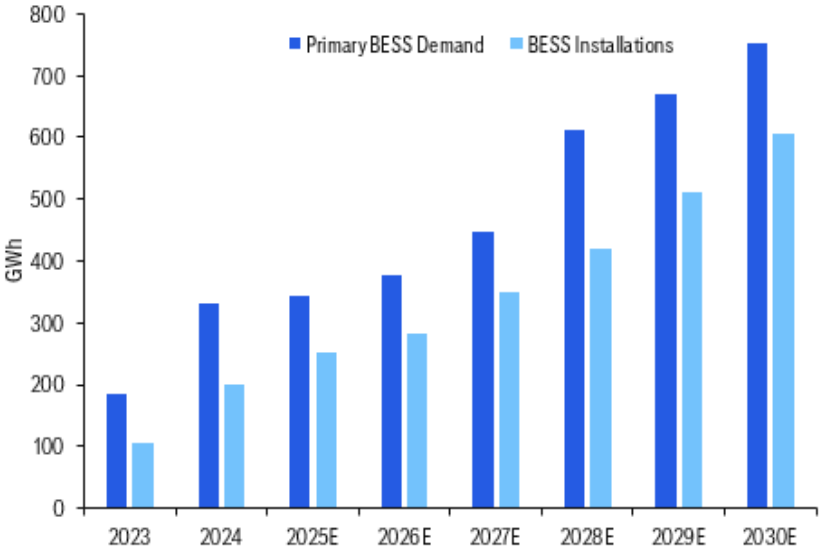


Renewables demand growth could stay strong, despite near-term headwinds

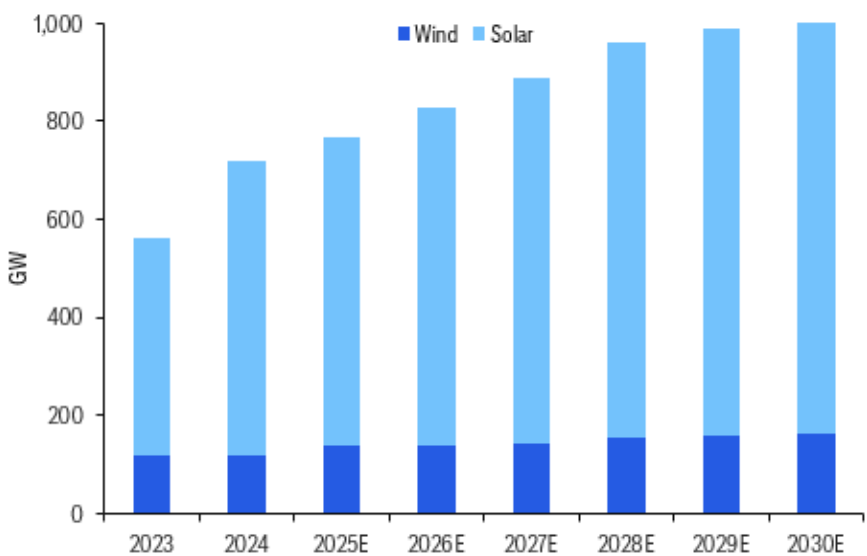
We see solar installations in China falling 2H'25 versus 1H'25 following policy-related frontloading. China's solar installations for the first four months of the year rose 75% y/y aided largely by front loading ahead of change in regulatory guidelines effective June 2025. Our China utilities team [expect](#) PRC solar installation to drop 39-44% h/h to 90-95GW^{ac} in 2H'25, from 155-160GW^{ac} in 1H'25 owing to the installation rush in 5M'25. The team has [revised](#) their global solar installation forecast to 535GW^{ac}, +0.9% y/y from 580GW^{ac}, +9% y/y mainly owing to less installations in China from 290GW^{ac} (+4.5% y/y) to 250GW^{ac} (-9.9% y/y). Despite slower growth in solar installations this year and the headwinds for 2H'25, we expect solar and wind demand growth to remain strong in the medium term.

Energy storage applications have underpinned demand for lithium (and copper to a lesser extent) over the last 12-18 months underpinned by strong wind and solar additions, particularly in China. We expect strong but slower growth of 27% this year due to changes in solar sector. Last year saw ~200GWh of battery ESS installed globally mainly paired with utility scale solar or wind installations. China ESS cell output has outpaced installations over the last few years aided by front loading from overseas markets ahead of potential tariffs (e.g. from US and EU). Despite China solar headwinds, we expect global BESS installations to grow ~27% y/y to 250GW.

In our base case, we expect primary BESS demand to touch ~800GWh by 2030



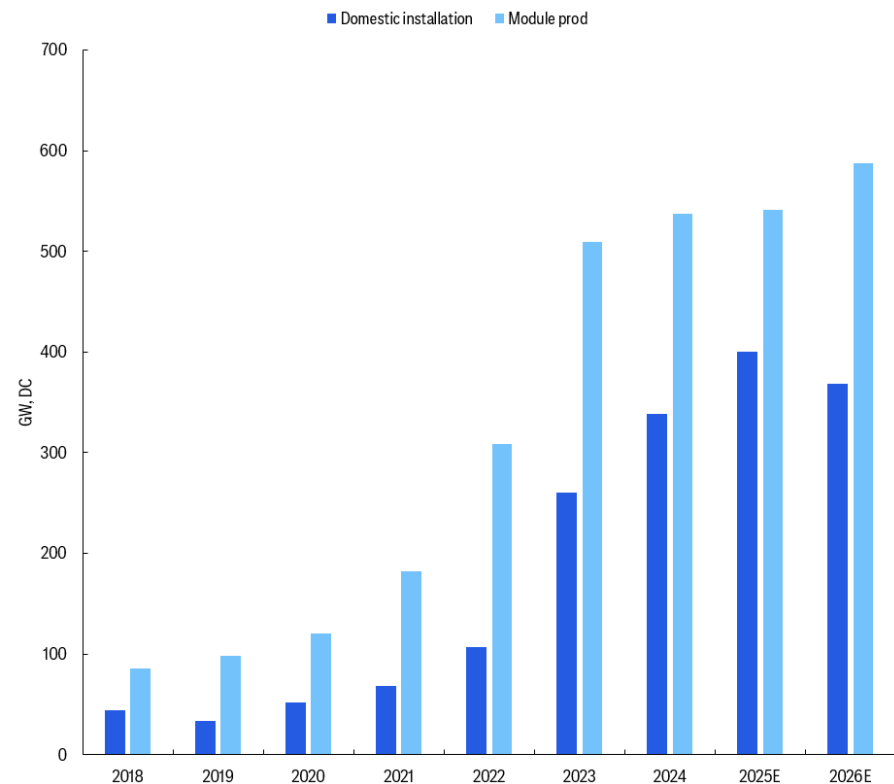
Citi solar and wind capacity addition forecasts* underpinning our metal demand projections (solar in GW/DC)



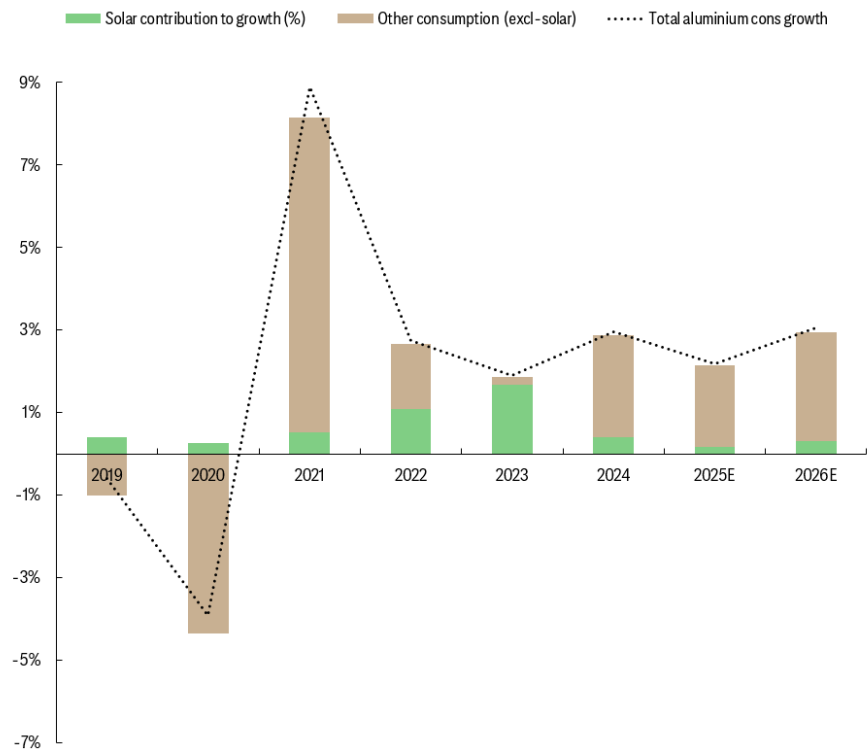
China solar installations slows, but module export keeps demand afloat

Our latest model update shows that the solar segment will continue to make a net positive contribution of around 0.2–0.3% to total aluminium demand growth in 2025–2026, down from 1.7% in 2023. Domestic installations have slowed sharply after policy-related front-loading earlier this year, while exports have remained resilient, sustaining overall demand across the solar value chain. This pattern is consistent with the view outlined in our [3Q25 Outlook](#) (p. 129) — ‘China solar installation slows, but module export keeps demand afloat’ — and the latest three months of customs data have validated this assumption.

China solar module production continue to rise even as domestic installation normalises



China solar sector remains a net positive contributor to aluminium demand growth, with export market strength deepening amid China’s overseas/Belt and Road penetration

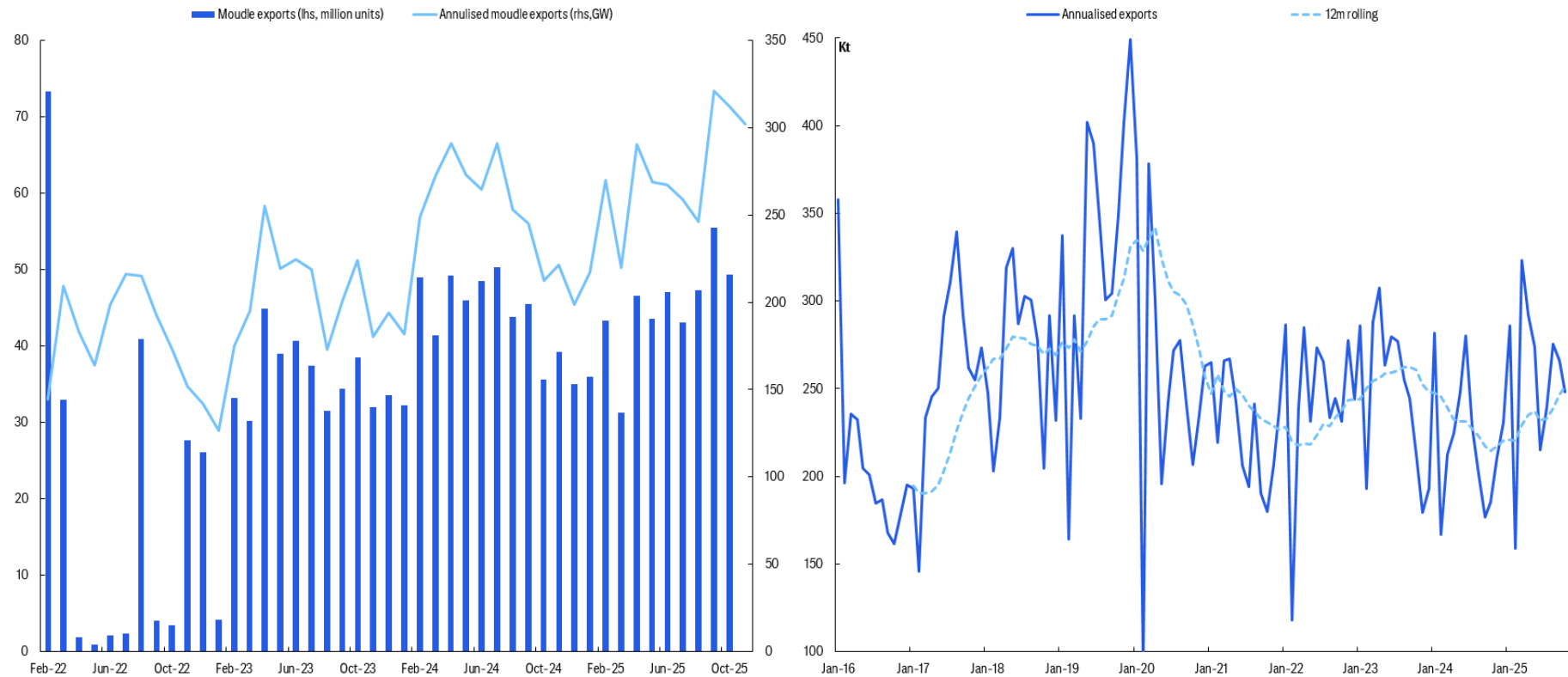


Multiple external drivers underpin China's solar value-chain growth

China's solar value chain—spanning modules, frames, and other components—are supported by a broad set of external drivers, with BRI-related projects build out providing a structural pull, Chinese manufactures accelerating their overseas expansion amid intensifying domestic competitions, and cost-competitive positioning of Chinese solar products remaining highly attractive to many emerging market economics.

China solar module exports remain resilient, sustaining >300GW (annualised) in 2025 amid a domestic installations slowdown

Aluminium frame (for solar modules) exports remain solid as China deepens its participation in the global solar value chain (>250kt annualised for 2025)



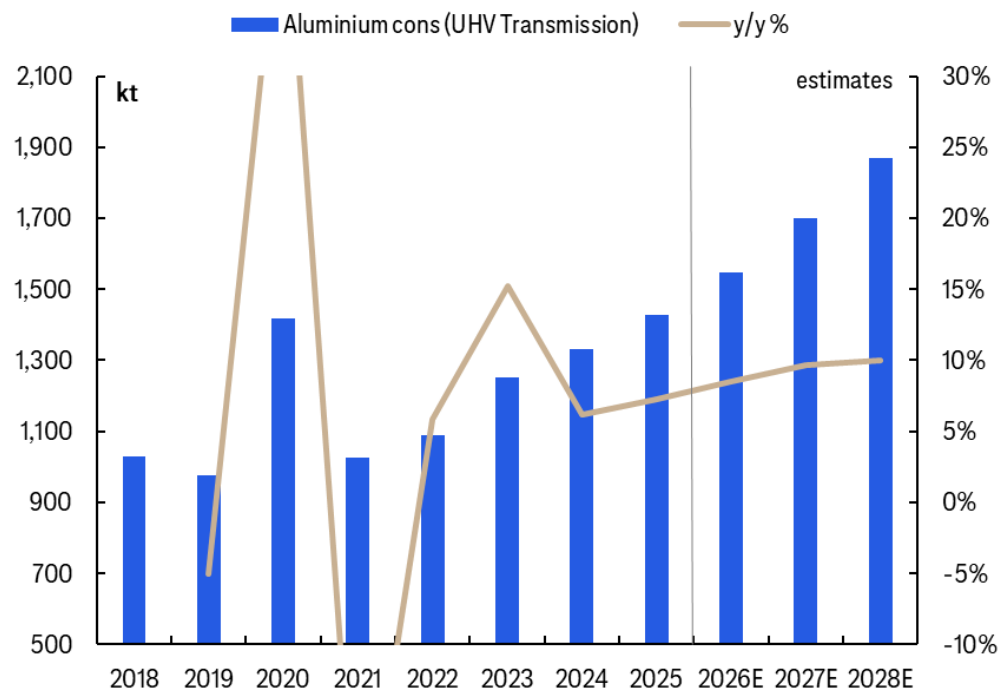
China's UHV and power equipment expansion catalysts of AL demand growth

Long-distance UHV (ultra-high-voltage) transmission projects - a core component of global power grid investment - are an important driver of aluminium demand. China's continued prioritisation of UHV expansion is expected to translate into incremental aluminium consumption in cables, highlighting how targeted investment within the transmission segment can support underlying demand growth.

Beyond individual projects, medium-term investment visibility also appears supportive. PRC power grid capital expenditure during the 15th Five-Year Plan period (2026–30E) is said to reach no less than Rmb4.1trn, implying an investment CAGR of c.5–6%. This provides a constructive backdrop for ongoing transmission and related equipment demand.

China's 2025–26 Work Plan for the power equipment industry (released September 2025) points to a broader policy push beyond transmission, with implications for downstream equipment demand. Given aluminium's widespread use across power conductors, solar module frames and equipment housings, the initiative is likely to provide an additional source of demand, reinforcing aluminium's role within the energy transition.

Modelled aluminium demand from State Grid's UHV spending - representing one segment of total transmission use with 7.5% y/y growth

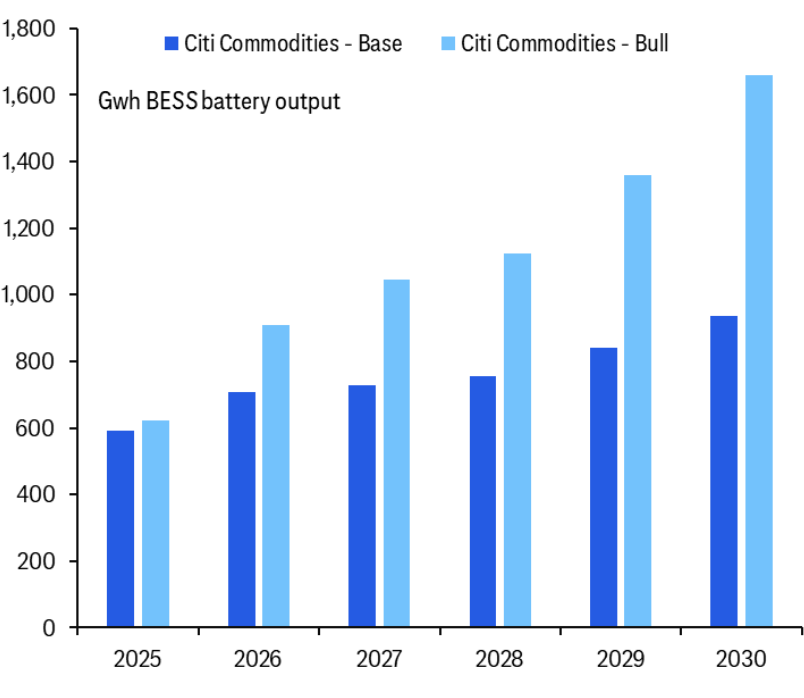


BESS – Material upside to metals consumption if bull case plays out

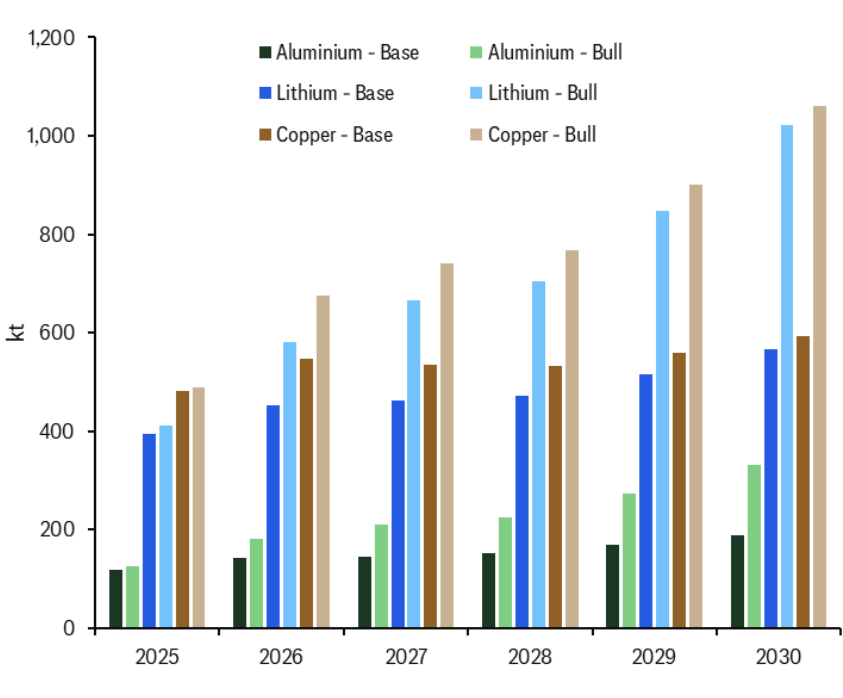
We see material upside to our demand estimates for aluminium, copper and lithium if our bull case BESS outlook thesis plays out. For lithium, we see material deficits warranting a higher price to incentivise supply and/or induce demand destruction. In our bull case scenario, we see BESS installations growing from 300GWh to ~1,200GWh by 2030 with output growing from 600GWh to ~1600GWh by 2030. Our China M&M team has forecasted BESS output to reach 1,200GWh by 2027 and if we extrapolate the growth trend, 2030 output could be closer to 2,000GWh. In the event the bull scenario plays, lithium will see a stockout situation as current growth pipeline may not be enough to satisfy the outlook (see lithium slides for bull case forecasts).

Besides lithium, copper and aluminium are beneficiaries of a stronger BESS outlook. For aluminium, we don't explicitly model separately for ESS batteries but we have estimated the implied aluminium consumption (below chart) based on the intensity of aluminium foil used in cells. If we include structural components such as aluminium shells and plates, the implied intensity will be higher. Copper demand reaches ~600kt by 2030 in our base case and ~1Mt by 2030 in our bull case.

Citi BESS output (base vs bull)



BESS metal consumption scenarios

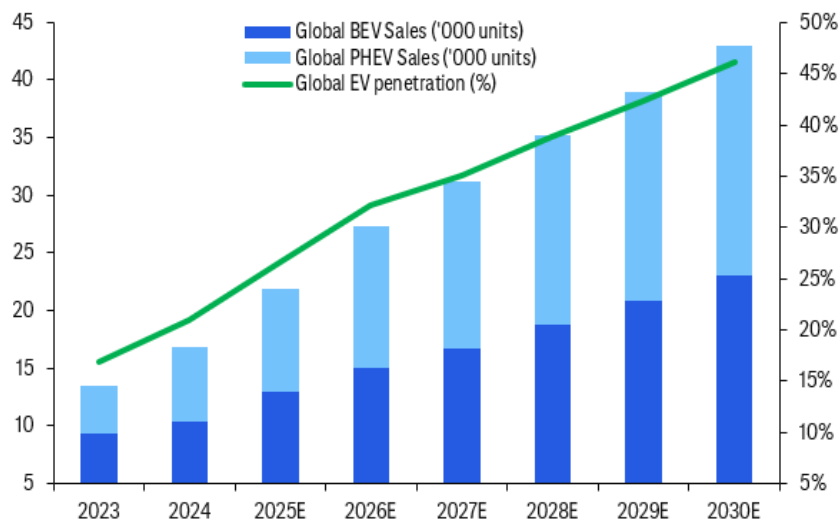


Global EV demand – China driving growth, US impacted by policy changes

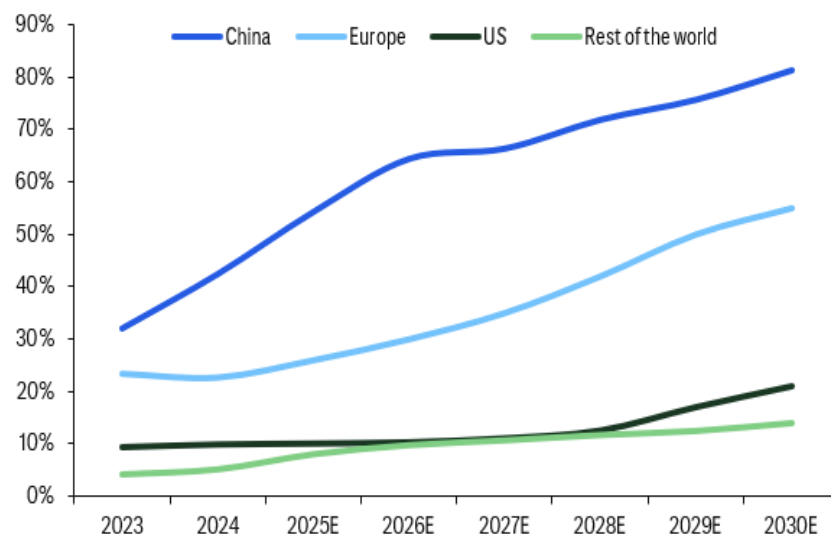
Citi expects global EV sales rising 30% y/y to ~22mn units in 2025, underpinned by ~40% y/y growth in China. Citi's China autos team recently upgraded their China wholesale EV forecasts, assuming strong growth in China EV (particularly PHEV) exports. They lift their global EV forecasts for CY25/26 by 3%/5%, underpinned by China EV sales. They are optimistic about growth in China EV sales, which rose 44% y/y in 4M'25, and see the trend continuing into 2H'25. However, rising price discounts have attracted increased regulatory scrutiny. National "trade-in" subsidies have reportedly been suspended in some provinces, earlier than the end of 2025 as previously expected ([Sohu.com](https://www.sohu.com), May 27, 2025). In addition, although China's exports are well diversified, exports to certain countries in future may attract regulatory headwinds and barriers, as well as competition from ex-China OEMs.

Policy change is likely to slow the pace of US EV adoption over the next 3-4 years. We see US EV penetration rates rising slowly to 20% by 2030 (previously c.30%). The proposed BBB bill, if approved by Senate, would largely phase out clean vehicle credits of \$7,500 by the end of 2026. However, the credit for most auto OEMs would phase out at the end of 2025 since credit into 2026 is only valid for auto OEMs that have sold >200k units. In addition, the new bill proposes a \$250 annual federal fee for EV owners. Large western auto OEMs are reworking their electrification strategy, with Honda planning to increase share of HEVs over pure battery electric vehicles (see [here](#)). HEVs generally have far smaller battery sizes compared to BEV/PHEVs with some using nickel metal hydride batteries rather than Li-ion cells.

We see global EV penetration reaching 45% by 2030



However, there are policy headwinds for western EV adoption



What challenges the bullish thesis?

What could challenge the bullish long-term outlook?

While our base case supports a structural bull market, several downside risks could derail the trajectory. These range from weaker demand recovery and unexpected supply surges in China and Indonesia, to faster scrap growth enabled by technological advances.

▪ Demand disappoints

Global cyclical weakness proves more persistent than expected, with macro headwinds (sluggish industrial activity, slower capex, weaker consumer spending), and rising government debt costs from higher long-end bond yields weighing on growth and delayed demand recovery

▪ China overproduces (China creep)

a. Creep production via amperage increases: Smelters raise output by boosting current load on existing pots (operationally possible but not sustainable).

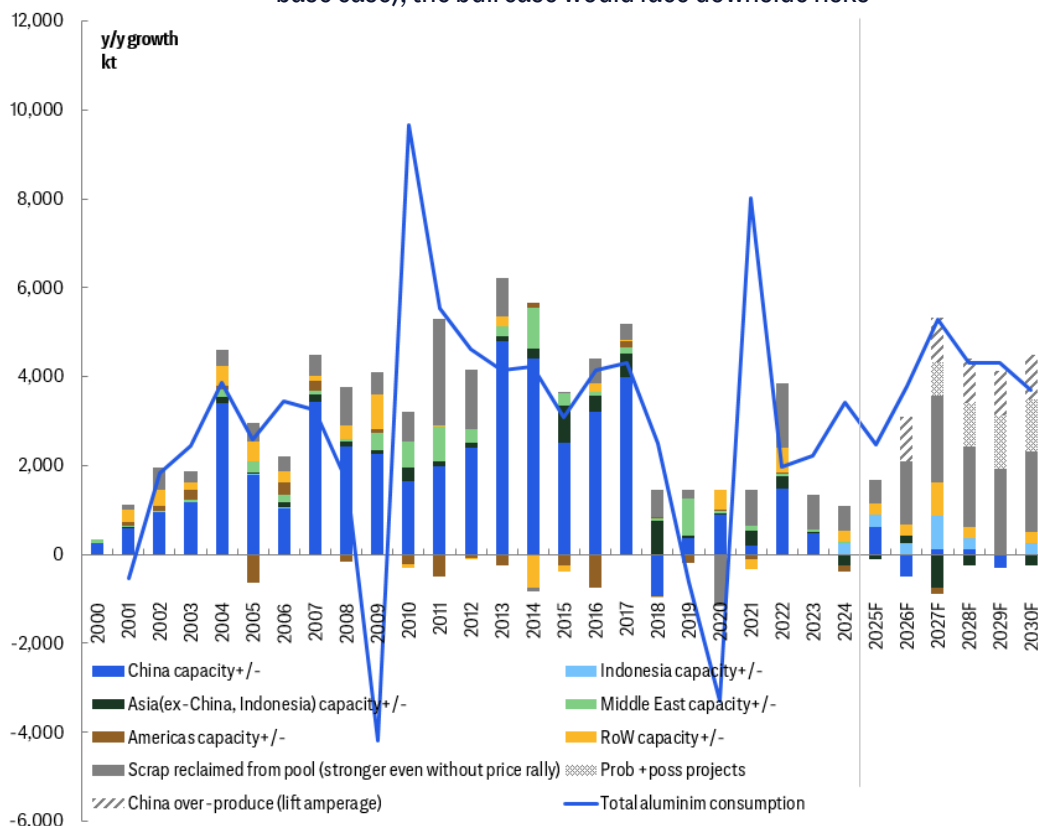
b. Breaking the cap: Beijing abandons or loosens the capacity ceiling, enabling new smelter additions (highly unlikely until we see US\$3,500–4,000/t aluminium price, in our view).

▪ Indonesia oversupplies

Indonesia brings on stream all currently known projects, plus additional unannounced capacity emerges “out of the woodwork”; faster-than-expected project execution, undermining the bull thesis.

▪ **Scrap supply growth accelerates without price response** (highly unlikely to be material in the base case for several reasons).

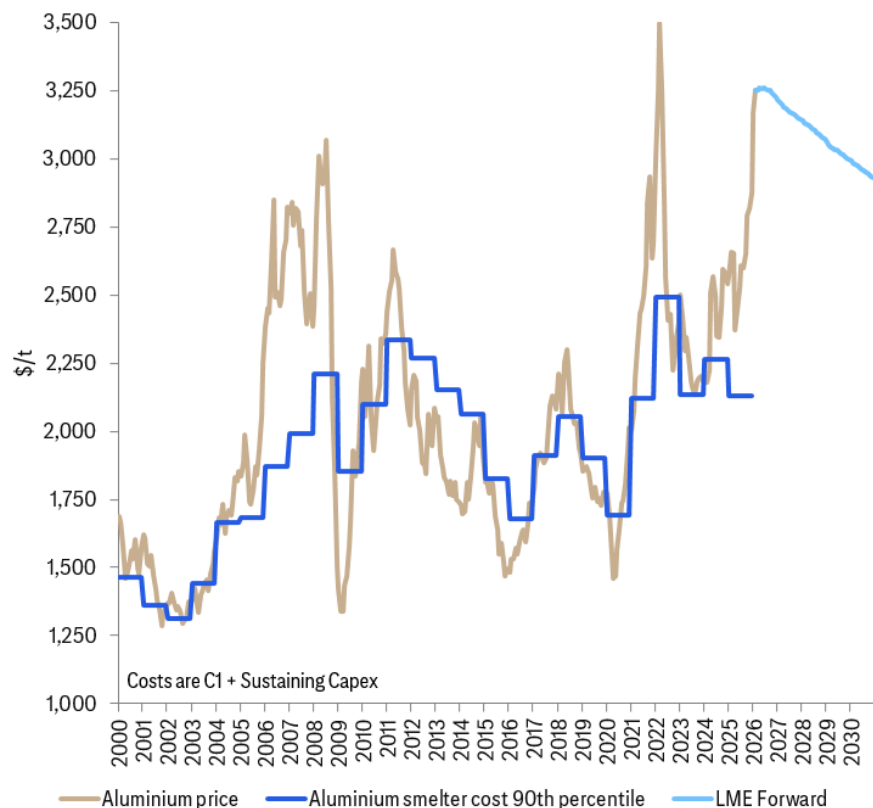
If China overproduces and scrap supply surges without price responses (not our base case), the bull case would face downside risks



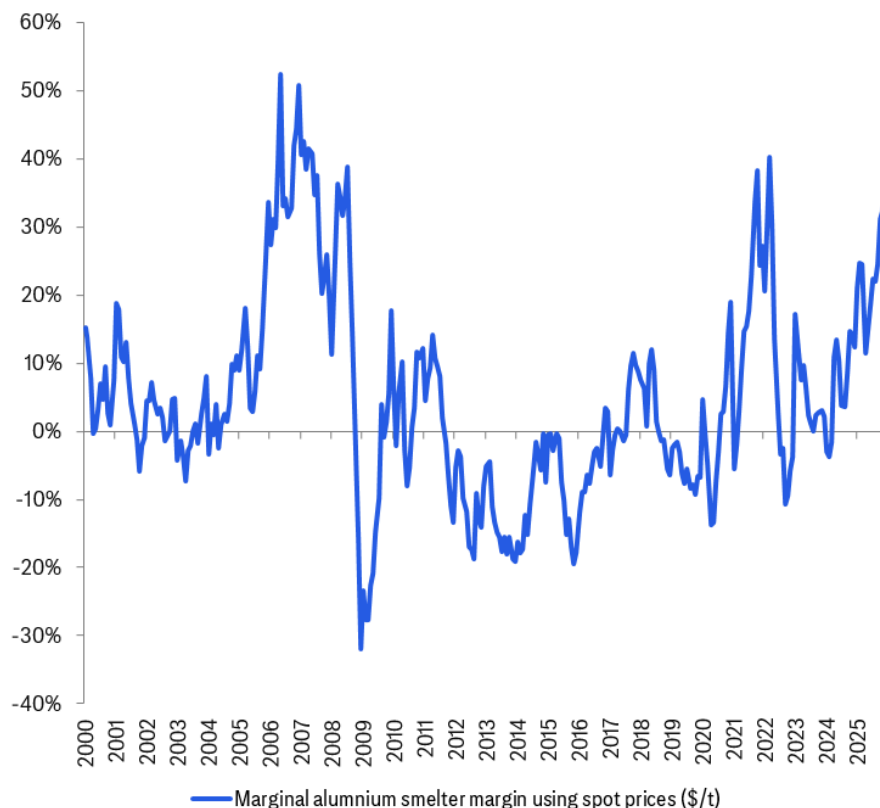
Corporate hedging considerations – insurance recommended

Aluminium smelters are enjoying margins nearing the 2006/7 and 2021 peaks

Aluminium prices have risen toward multi-year highs and have disconnected from marginal smelter costs, with forward prices now well above spot



Marginal aluminium smelter margins are the strongest in years, reaching levels not seen since the early-2000s



Commodity insurance and the Mexican oil hedge program (1)

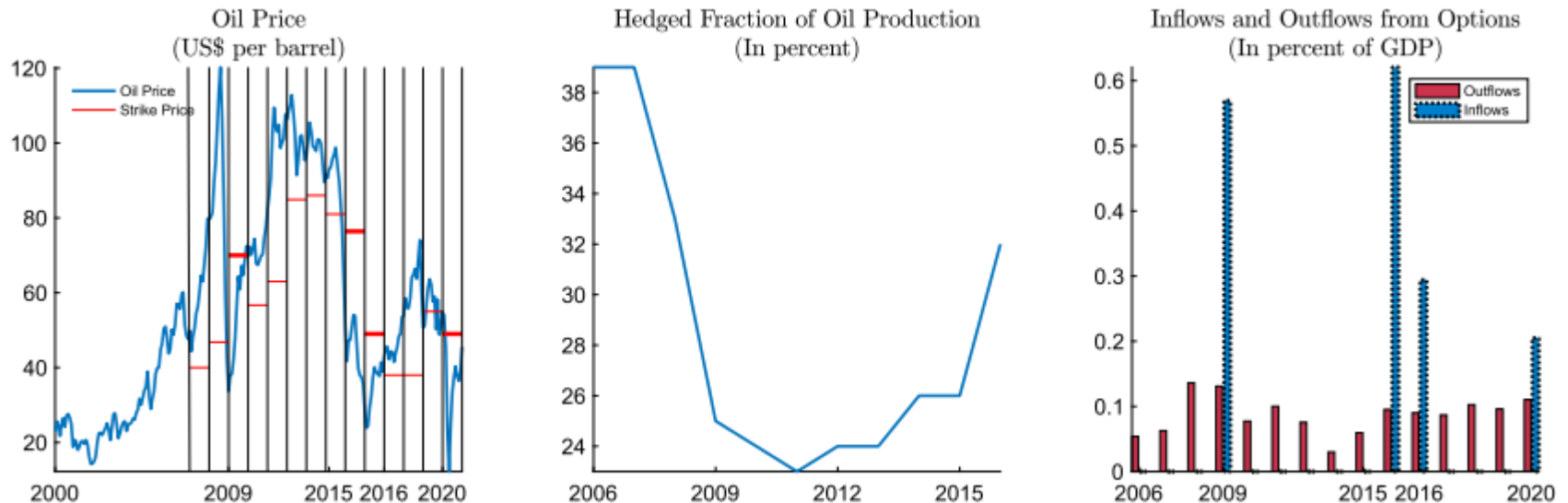
We continue to believe that aluminium consumers should insure against higher prices. We do not recommend aluminium producers cap their upside, rather we would recommend insurance – as the risks to the global economy/geopolitics are evident in gold's price boom.

Indeed, a review of the history of corporate and sovereign hedging problems indicates that most of the issues that occurred were on the producer side, and reflected the selling of upside, and even more specifically, the overselling price upside. The overselling of upside can leave sovereigns and/or companies vulnerable to unanticipated cost inflation, production underperformance, reduced equity valuation due to some shareholders' desire to have exposure to upside, and in some cases, cash margin calls.

According to a recent paper in the *Journal of International Money and Finance*, Mexico gained the benefits of hedging – lower revenue volatility, with no major hedging issues over the past 20 years, through the use of insurance. They did not sell upside, by paying on a rolling basis an insurance premium. The benefits for Mexico have been dramatically lower earnings volatility, lower costs of sovereign debt, and welfare gains, according to IMF and academic studies. The corporate equivalent (mechanically/theoretically) would be lower earnings and dividend volatility, lower beta, lower WACC, and higher equity valuation.

Of course, this comes at a cost, as insurance isn't free. Mechanically, some of this insurance premium cost can be offset by selling call spreads, meaning the selling of upside is finite/limited.

Mexico's oil revenue insurance (hedging) program



Note. Data is from INEGI. Mexican government stopped publishing the number of barrels hedged after 2016.



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Welfare gains from market insurance: The case of Mexican oil price risk ☆

Chang Ma ^{a,*}, Fabián Valencia ^b

^a Fudan University (FISF), Room 514, No. 5 Building, 220 Handan Rd., Shanghai 200433, China

^b International Monetary Fund, 700 19th St NW, Washington, DC 20431, United States of America

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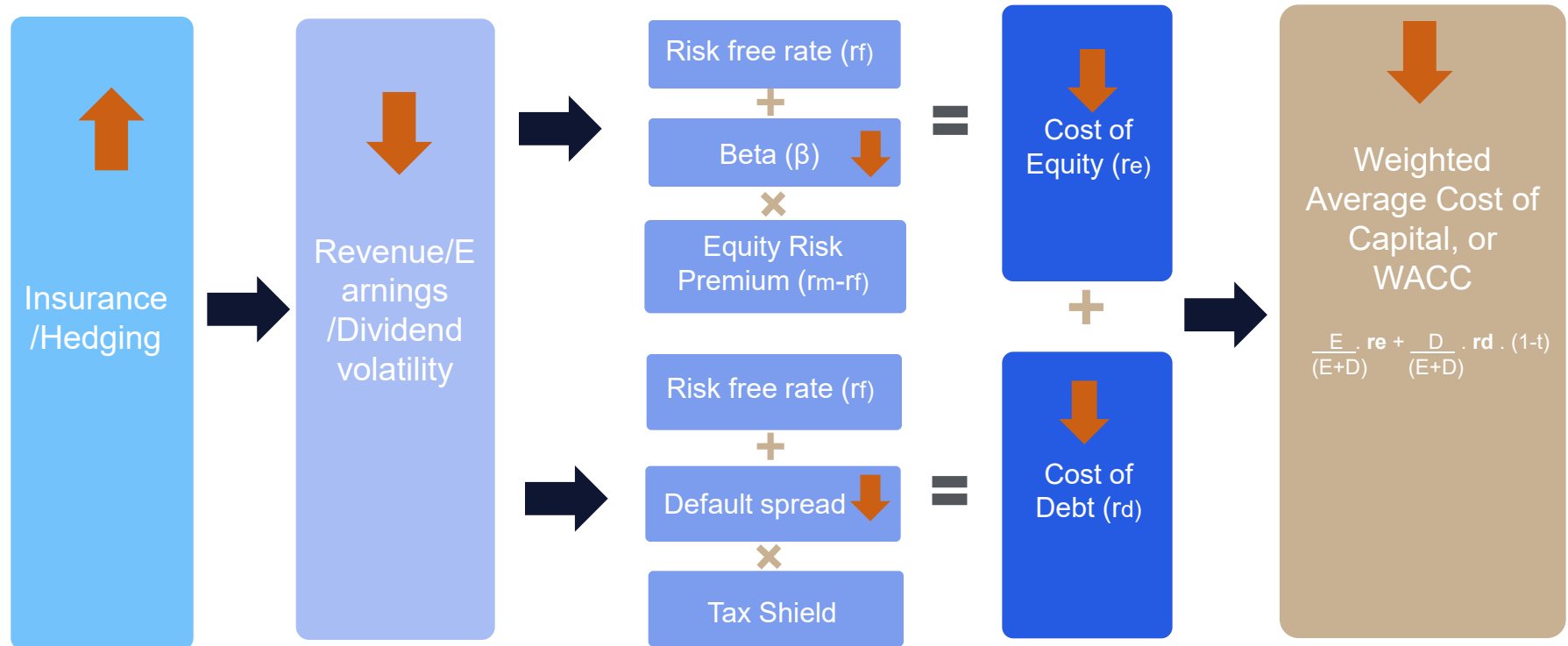
Default

ABSTRACT

Mexico has a long-standing practice of hedging oil price risk through the purchase of put options. We examine the resulting welfare gains using a standard sovereign default model calibrated to Mexican data. We show that hedging increases welfare by reducing income volatility and reducing risk spreads on sovereign debt. We find welfare gains equivalent to a permanent increase in consumption of 0.44 percent with 90 percent of these gains stemming from lower risk spreads.

The value of insurance comes through lowering earnings/dividend volatility

Hedging can break the link between commodity volatility and earnings volatility, reducing the probability of default on debt and losses on equity, lowering a firm's cost of debt and/or equity capital, and driving up firm value, all else equal. A firm may attempt to increase their valuation through hedging, which protects earnings in a downturn and allows for more stable cash flows. Capital Asset Pricing Theory states that the proper measure of risk of a security is the systemic risk, commonly referred to as beta of a security. The firm's beta is a measure of a stock's volatility in relation to the market. An increase in beta (increase in earnings volatility) increases the cost of equity through the Capital Asset Pricing Model (CAPM).

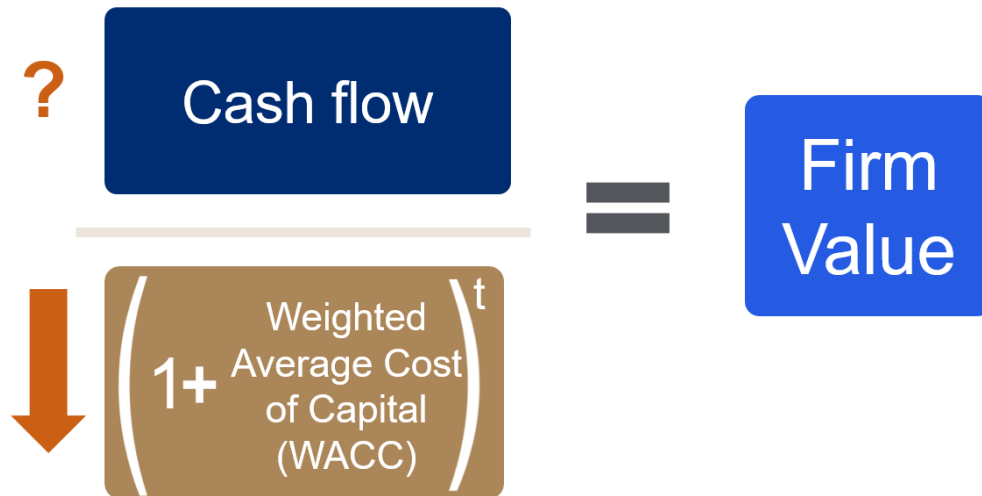


Where: E = Market value of the firm's equity; D = Market value of the firm's debt; r_e = Cost of equity; r_d = Cost of debt; t = Corporate tax rate

The cost is that insurance/hedging has an uncertain impact on cash flows

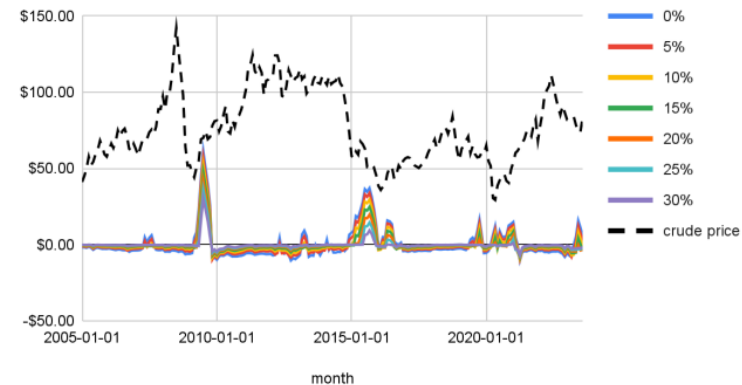
Over the short and long run hedging / insurance has an uncertain impact on cash flows. The Mexico oil hedging program has both lowered volatility and cost nothing over the long run. For gold, this would not have been true, given gold has risen from \$300/oz to \$4000/oz over the over the past 25 years. Over short run periods, when oil prices were high / well above costs of production, and prices subsequently fell, the cash flows were positive (2009/2015/2020), and negative during the interim periods.

According to DCF valuation

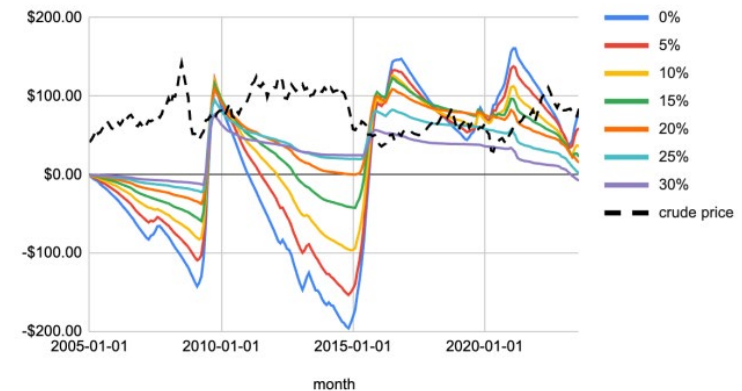


Mexico's oil revenue insurance program estimates of potential costs, payouts, at various strike discounts

Hedging Profits At Various Strike Discounts



Cumulative Profits at Various Strike Discounts





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Appendix A-1

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Citigroup Global Markets India Private Limited

Shreyas Madabushi

Citigroup Global Markets Limited

Wenyu Yao; Maximilian J Layton

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